

Does Liquidity Management Induce Fragility in Treasury Prices? Evidence from Bond Mutual Funds

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Mutual funds investing in illiquid corporate bonds actively manage Treasury positions to buffer redemption shocks. This liquidity management practice can transmit non-fundamental fund flow shocks onto Treasuries, generating excess return volatility. Consistent with this hypothesis, we find that Treasury excess return volatility is positively

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associated with bond fund ownership, and this pattern is more pronounced among funds conducting intensive liquidity management. Causal evidence is provided by exploiting the U.S. Securities and Exchange Commission's 2017 Liquidity Risk Management Rule. Evidence also suggests that the COVID-19 Treasury market turmoil was attributed to intensified liquidity management, an unintended consequence of the 2017 Liquidity Risk Management Rule. (*JEL* G01, G12, G14, G23)

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While investors conventionally view the U.S. Treasury market as a safe haven, regulators have concerns about the increasing fragility of the Treasury market. In 2016, Jerome Powell, the current chair of the Federal Reserve, pointed out that “spikes in volatility and sudden declines in liquidity have become more frequent in both Treasury and equity markets ... [t]here is also evidence that liquidity shifts more rapidly and hence is less predictable in these markets” (Powell 2016). Several recent episodes in the Treasury market exemplify this statement, including the “taper tantrum” in 2013, the “flash rally” in 2014, and the COVID-19 turmoil in March 2020.¹ It is not completely clear what economic mechanism drives increasing fragility in the Treasury market—the most liquid market in the world.

In this paper, we argue that the common practice of liquidity management among open-end corporate bond mutual funds contributes to increased fragility in the Treasury market. Mutual funds that invest in corporate bonds (or other illiquid assets) perform liquidity transformation—holding illiquid securities but issuing liquid claims (i.e., fund shares) to investors. These funds often face run risks arising from strategic complementarities among investors (e.g., Chen, Goldstein, and Jiang 2010; Goldstein, Jiang, and Ng 2017). To mitigate the risk, bond mutual funds actively engage in liquidity management, that is, they maintain a large amount of cash-like or highly liquid assets as a buffer for investor withdrawals (see Choi et al. 2020; Jiang, Li, and Wang 2021). As a result, their tradings on liquid assets—mostly Treasuries—appear to be excessively sensitive to investors' demands for liquid claims. We argue that this liquidity management can potentially transmit and concentrate the nonfundamental shocks, which are driven by fund flows, onto the price of Treasuries held by the funds, generating excessive volatility in the Treasury market. Such an effect

¹ For discussion of the “taper tantrum,” see Adrian et al. (2015); for discussion of the “flash rally,” see Joint Staff Report (2016); for discussion of Treasury market performance in March 2020, see Duffie (2020), Fleming and Ruela (2020), He, Nagel, and Song (2022), Schrimpf, Shin, and Sushko (2020), and the U.S. Federal Reserve's Financial Stability Report (2020).

has been particularly pronounced in recent years as the total size of open-end funds that invest in illiquid assets has increased manifold.²

To test the asset pricing implications of liquidity management on Treasuries, we focus on global open-end corporate bond mutual funds (for brevity, labeled as “bond funds” hereafter) from 2002 through 2021.³ Corporate bond funds are ideal for testing our argument because they have several unique features. First, bond funds usually trade two major asset classes with distinct liquidity levels: U.S. Treasuries and corporate bonds. Second, detailed data on fund holdings is available at the quarterly frequency for a long period, which allows us to analyze funds’ trading behaviors directly. Third, we can measure investors’ demand for liquid claims precisely by fund flows. Last, including global corporate bond funds is important when we exploit the 2017 Liquidity Risk Management Rule to pin down the causal impact of liquidity management.

Before testing the asset pricing implications, we first confirm our premise that bond funds indeed use U.S. Treasuries as a buffer to manage liquidity. Specifically, we examine whether bond funds disproportionately adjust their holdings of Treasuries and corporate bonds in response to fund flows. We find that, for example, with a 1% fund inflow, funds increase their holdings in U.S. Treasuries by about 1.42%, but increase their holdings in corporate bonds by only 0.84%. Moreover, the excessive purchase of Treasuries to contemporaneous fund flow tends to revert in the subsequent quarter. To corroborate the evidence, we take advantage of the global coverage of our bond fund holding data and further examine the trade-to-flow sensitivity of other government bonds.⁴ We find that the trade-to-flow sensitivity of other government bonds is 1.18, which is between that of U.S. Treasuries and corporate bonds. This pecking order, as suggested by trade-to-flow sensitivities, is consistent with liquidity management and increases in asset liquidity.

Then, we examine how liquidity management affects the prices of Treasuries, which are arguably the most liquid assets in the world. As bond funds aggressively trade Treasuries as a liquidity buffer in response to fund flows, this could generate strong price pressure on Treasuries. The fund flow–induced price pressure can cause excessive price fluctuations in Treasuries with high bond fund ownership (see argument in, e.g., [Greenwood and Thesmar 2011](#)). Moreover, such exposure becomes a defining factor for Treasury prices

² According to the [Investment Company Institute \(2022\)](#), total assets under management of open-end mutual funds with primary investment in illiquid assets—such as corporate bonds, municipal bonds, and bank loans—increased from 1.3 trillion USD in 2002 to about 7.3 trillion in 2019.

³ We exclude government bond funds, bank loan funds, and money market funds to avoid potential mechanical effects. For example, the impact of the ownership from government bond funds can arise from flow-induced trades rather than from liquidity management. [Appendix Table A2](#) provides details on bond fund sample construction. Our main results are also robust to the sample of only U.S. open-end corporate bond mutual funds.

⁴ To obtain these government bonds, we first identify government-related holdings from Morningstar categories and then exclude U.S. Treasuries. Most other government bonds in our sample are issued by developed countries.

as mutual funds are now a major participant in the Treasury market. [Liang \(2020\)](#) estimates that the marketable Treasury shares held by long-term mutual funds increased from 3% in 2008 to 8% in 2019—more than the amount held by banks and broker-dealers, and that bond funds trade much more actively than other major market participants, such as pension funds and insurance companies.

To this end, we first hypothesize that there should be a positive association between the excess return volatility of Treasuries and bond fund ownership.⁵ We exploit the cross-sectional variations of Treasuries' bond fund ownership; these variations can rule out confounding effects from the time series and better pin down the underlying economic mechanism. Excess volatility is calculated as the standard deviation of Treasuries' daily risk-adjusted returns, which are the residuals from either a market model that adjusts for average returns on Treasuries (following [Frazzini and Pedersen 2014](#)) or a three-factor model that adjusts for average returns on Treasuries, investment-grade corporate bonds, and junk bonds (following [Choi, Kronlund, and Oh 2022](#)).⁶

To examine our conjecture, we conduct Fama-Macbeth regressions controlling for Treasuries characteristics, including time-to-maturity, coupon rate, on-the-run status, bid-ask spread, and issue size. We indeed find that Treasuries' excess return volatility is positively associated with bond fund ownership. Such an effect is statistically significant and economically meaningful. We find that a one-standard-deviation increase in bond fund ownership is associated with a 0.0029% higher volatility of market-adjusted daily returns. For reference, during our sample period from 2002 through 2021, the average market-adjusted excess return volatility of Treasuries was 0.065%, so the effect is about 4.4% relative to our sample mean.⁷

Next, we link liquidity management directly to Treasury excess return volatility. To gauge the extent to which a fund uses Treasury positions to buffer flow shocks, we first construct a fund-level measure called liquidity management intensity (LMI). At each quarter, LMI equals the coefficient estimate from a univariate regression of a fund's net purchase of U.S. Treasuries

⁵ Thanks to an anonymous referee for suggesting that we focus on Treasury excess return volatility—a direct and natural measure of fragility. To our knowledge, we are the first to study the effect of liquidity management on Treasury excess return volatility. Meanwhile, the cross-sectional tests using a two-decade-long sample period that covers both normal and crisis periods can help us tease out confounding factors and have implications beyond stress periods.

⁶ Including corporate bond benchmark returns is to control risk factors in fund returns, which potentially affect fund flows and then Treasury returns. We also find similar results by including the additional term-structure of government bond yields ([Chen, Ferson, and Peters 2010](#)) in calculating excess return volatility, as shown in [Appendix Table A6](#).

⁷ Put differently, a 10% increase in mutual fund ownership is associated with a rise in Treasury return volatility of about 11% of the sample mean. For a comparison, [Greenwood and Thesmar \(2011\)](#) find that a 10% increase in mutual fund ownership leads to an increase in daily stock volatility of about 10% of their sample mean. This comparison suggests that the effect of mutual fund ownership on Treasury return volatility is economically meaningful, as it is comparable to, if not greater than, the effect of mutual fund ownership on stock return volatility.

on contemporaneous fund flows over the previous 12 quarters. Intuitively, when one bond fund conducts more liquidity management with Treasuries, its LMI is greater. Consistent with the results on the trade-to-flow sensitivity on the Treasuries, the average LMI across funds is greater than 1 (equals 1.25). In this way, LMI can capture and take into account all prior factors that motivate liquidity management, such as cash holdings, portfolio liquidity, and fund flow volatility (Chernenko and Sunderam 2016; Jiang, Li, and Wang 2021). We empirically confirm that this is the case in the data: funds with low cash holding, high flow volatility, and high fund illiquidity tend to exhibit strong LMI.

We then split our sample of bond funds based on liquidity management intensity in each quarter: the top one-third are high-LMI funds, whereas the rest are low-LMI funds. We hypothesize that ownership of high-LMI funds induces stronger excess volatility of Treasury returns than does ownership of low-LMI funds. The evidence confirms this conjecture: the effect of ownership on volatility for high-LMI funds is more than four times that for low-LMI funds.⁸ Meanwhile, we confirm that different proportions of Treasury holdings in fund portfolios do not drive the differential effects of high- and low-LMI fund ownership. We further show that our results are robust after excluding the recent COVID-19 crisis period (i.e., 2020Q1 and 2020Q2), suggesting that our findings can be generalized beyond the COVID-19 Treasury market turmoil.

While the above findings are consistent with our argument that liquidity management of bond funds induces excess volatility in Treasuries, we are aware of potential endogeneity issues. For example, bond funds may prefer to hold Treasuries with certain characteristics that could lead to higher excess return volatility in the future. Fund characteristics (such as fund performance, particularly during the stress periods) can potentially endogenously determine fund trading behavior.⁹ While such an argument is unlikely from the perspective of liquidity management, we exploit two empirical designs to provide evidence for a causal interpretation.

In our first empirical design, we exploit the effect of the U.S. Securities and Exchange Commission's (SEC) Liquidity Risk Management Rule adopted in 2017. The Rule sets a minimum requirement for liquid assets for U.S. mutual funds. Such a rule forces funds investing in illiquid corporate bonds to conduct liquidity management with liquid assets; the impact of this policy should be more pronounced for U.S. funds that do not actively manage liquidity beforehand. We use this policy and conduct a difference-in-differences (DiD)

⁸ Likewise, we find that the positive association between Treasury excess return volatility and fund ownership is stronger among funds with severe liquidity mismatch. Liquidity mismatch refers to the difference between the portfolio weight of corporate bonds minus the sum of portfolio weights in cash and government bond holdings. See [Appendix Table A11](#).

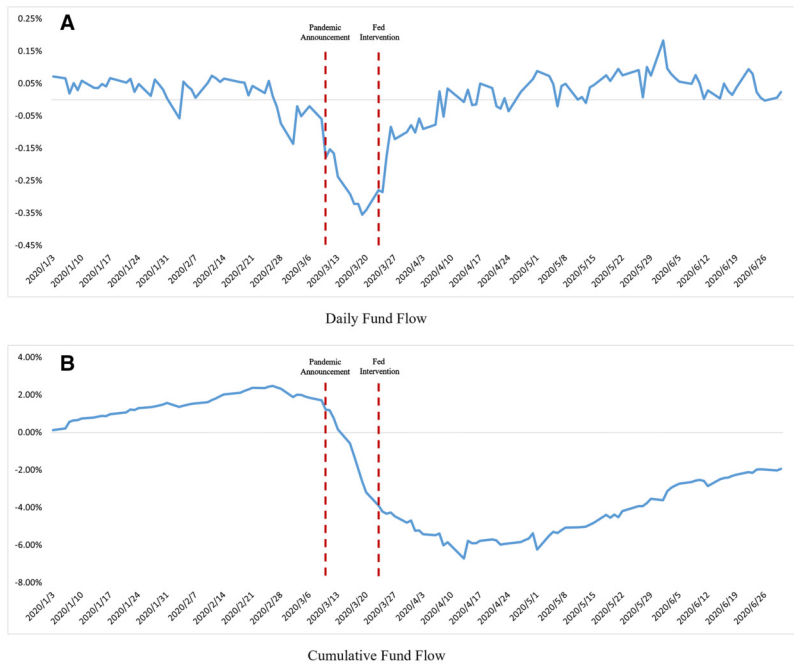
⁹ We follow the suggestion from one anonymous reviewer and consider an alternative measure of bond fund ownership—the residual fund ownership from regressing on lagged Treasury return volatility. We confirm that our results are robust and are not driven by the lagged Treasury return volatility. Results are reported in [Appendix Table A10](#).

analysis. In the DiD analysis, the treatment group consists of low-LMI U.S. funds, and the control group consists of other funds, including high-LMI U.S. funds and all non-U.S. funds. We first verify our premise that low-LMI U.S. funds conduct more liquidity management with Treasuries after the policy is implemented. Specifically, low LMI U.S. funds significantly increase their trade-to-flow sensitivity for Treasuries. In contrast, other funds do not appear to increase their liquidity management intensity after the rule. Furthermore, we examine the effect on Treasuries' volatility. We find that Treasuries heavily held by these low-LMI U.S. funds experience an increase in excess volatility in the post-event period, consistent with the increased liquidity management with Treasuries of these funds after introducing the Liquidity Risk Management Rule in 2017.

The second empirical design is based on the turmoil in the Treasury market in March 2020. Bond mutual funds experienced large fund outflows during the COVID-19 outbreak period; this provides an ideal opportunity to study whether liquidity management induced price declines in Treasuries. Consistent with prior studies (e.g., [Falato, Goldstein, and Hortaçsu 2021](#); [Ma, Xiao, and Zeng 2022](#)), from the beginning of March 2020 (shortly before the World Health Organization announced the global pandemic), bond funds experienced significant fund outflows until the market was stabilized with the Fed's intervention and bailout (see Figure 1). In our sample, the total fund outflow during this period was more than 5% of bond funds' pre-event total assets under management (AUM). More importantly, we find that Treasuries heavily held by high-LMI funds (as measured before the event) experienced larger price declines during this period. Also, such price declines were fully reversed after the Fed's intervention, suggesting that the price declines during the COVID period were not fundamentally driven but arose from liquidity-management-induced price pressure.¹⁰

The two tests based on the 2017 Liquidity Risk Management Rule and the COVID-19 market turmoil not only buttress the identification but also offer new insights into the puzzling pattern of Treasury prices during the COVID-19 crisis. As we know, during periods of financial market turmoil, the prices of Treasuries usually increase due to flight-to-safety and flight-to-liquidity. The COVID-19 crisis, however, was dramatically different, as Treasuries experienced sharp price declines. This phenomenon is puzzling to academics and practitioners, raising concerns about the safe-haven status of U.S. Treasuries (see, e.g., [Duffie 2020](#); [He, Nagel, and Song 2022](#)). Our results suggest that this can be an unintended consequence of the 2017 Liquidity

¹⁰ [Ma, Xiao, and Zeng \(2022\)](#) also examine the COVID-19 periods and find that outflow-induced trades generate selling pressures on off-run Treasuries. However, the asset pricing implication of liquidity management during the COVID-19 crisis is unclear and not directly tested in their study. The differential effect of high- and low-LMI funds on Treasury price declines in our study clearly confirms that liquidity management was one of the driving forces of the sharp price decline in Treasuries during the COVID-19 turmoil, complementing the work of [Ma, Xiao, and Zeng \(2022\)](#).

**Figure 1****Fund flows during COVID-19 pandemic announcement and Fed intervention**

This figure plots the weighted average daily fund flows (panel A) and cumulative fund flows (panel B) for all bond funds in our sample from 2020Q1 through 2020Q2.

Risk Management Rule, as the policy intensified liquidity management of bond funds, concentrating flow shocks onto the price of Treasuries with the COVID-19 outbreak.¹¹

We further extend our study and examine the return comovement of Treasuries for completeness. Return correlations among individual assets within a particular asset class largely determine the total return variance of the asset class, which is another natural measure of fragility. Thus, the study of return comovement of Treasuries can further shed light on the systematic risk in the Treasury market. Specifically, we follow [Anton and Polk \(2014\)](#) to calculate common ownership of bond funds for each pair of Treasuries and find that common ownership positively forecasts comovement among Treasuries, consistent with the studies of [Greenwood and Thesmar \(2011\)](#) and [Anton and Polk \(2014\)](#) on equities. More relevant for market fragility, we

¹¹ Echoing this point, we further investigate the implications of the Secondary Market Corporate Credit Facility (SMCCF), a policy from the Federal Reserve to boost the liquidity of corporate bonds. We find that Treasuries held heavily by bond funds with high exposure to SMCCF-eligible bonds experienced less excess return volatility following the commencement of the SMCCF. This result, together with the result on the 2017 Liquidity Risk Management Rule, sheds light on how the Federal Reserve can mitigate Treasury fragility in the future.

show that the association between common ownership and the Treasury return comovement is stronger during downside markets than during upside ones. On the contrary, there is no such asymmetric pattern on corporate bonds. These findings highlight the uniqueness of Treasuries in liquidity management and are consistent with the intuition on the urgency of selling Treasuries to meet fund redemption during downside markets.

Related Literature. Our study contributes to several strands of literature. First, it is closely related to the growing literature on financial fragility and liquidity management of mutual funds. When mutual funds perform liquidity transformation—holding illiquid assets but issuing liquid claims to investors—they are often subject to financial fragility due to strategic complementarities among investors (e.g., [Chen, Goldstein, and Jiang 2010](#); [Goldstein, Jiang, and Ng 2017](#)); this was particularly true during the COVID-19 pandemic (for detailed evidence, see [Falato, Goldstein, and Hortaçsu 2021](#)). To mitigate this financial fragility, mutual funds use cash or cash-like assets to manage their liquidity needs (see [Chernenko and Sunderam 2016](#); [Aragon, Ergun, and Girardi 2022](#); [Jiang, Li, and Wang 2021](#)). In addition, [Jotikasthira, Lundblad, and Ramadorai \(2012\)](#) show that emerging market funds prefer to trade assets in more liquid markets when accommodating fund flow shocks. The main focus of these studies is to identify the strategic complementarities among investors or to document the liquidity management of mutual funds. Our study incrementally extends this literature by systematically studying how global corporate bond mutual funds trade different asset classes in response to fund flows and by examining the pricing impact of liquidity management on the buffer assets, that is, Treasuries. We show that liquidity management can induce sizable price pressure and volatility among Treasuries. Our finding based on the SEC's Liquidity Risk Management Rule of 2017 is also novel to the literature, as it helps to pin down the causal impact of liquidity management on the excess volatility of Treasuries.¹²

[Ma, Xiao, and Zeng \(2022\)](#) is the contemporaneous work most related to our study. While we share a similar motivation (liquidity management with Treasuries), our paper has different focuses. [Ma, Xiao, and Zeng \(2022\)](#) look solely at the COVID-19 period, documenting the liquidation behavior of various types of fixed-income mutual funds and showing evidence of price pressure from imputed fund outflow. By comparison, our study focuses on a more general setting by using a direct measure of fragility (excess return volatility) and a longer sample period (including both normal and crisis times). Our study has at least three advantages. First, while [Ma, Xiao, and Zeng \(2022\)](#)

¹² Our findings on Treasuries are also related to [Greenwood and Vayanos \(2010\)](#), who present anecdotal evidence from two events (the U.K. pension reform of 2004 and the U.S. Treasury's buyback program of 2000–2001) to show that the government bond market can be affected by short-term price pressures. Our findings on corporate bonds also echo those of [Choi et al. \(2020\)](#), who show that, due to the practice of liquidity management, flow shocks have little effect on corporate bond prices.

find that outflow-induced trades generate selling pressures on Treasuries, the role of liquidity management during the COVID-19 crisis is unclear. The differential impacts of high- and low-LMI bond funds on the price declines among Treasuries during the COVID-19 period clearly show the role of liquidity management, complementing the work of [Ma, Xiao, and Zeng \(2022\)](#). Second, one of our important innovations is exploiting the 2017 Liquidity Risk Management Rule as a quasi-natural experiment to pin down the causal impact of liquidity management on Treasuries. One of the biggest challenges to the literature on liquidity management is to find an exogenous shock to liquidity management and then identify its causal effect on asset prices. We push forward the literature on liquidity management by providing evidence for causal interpretation. Third, our findings based on the 2017 Liquidity Risk Management Rule and the COVID-19 period suggest that the COVID-19 Treasury market turmoil may have been an unintended consequence of the 2017 Liquidity Risk Management Rule, shedding light on the puzzling and unconventional Treasury market during the COVID-19 period (see, [Duffie 2020](#); [He, Nagel, and Song 2022](#)).

Our study also has policy implications and sheds light on the debate over liquidity management policies. The extension to the Secondary Market Corporate Credit Facility, a policy introduced by the Federal Reserve to boost the liquidity of corporate bonds, together with our study on the 2017 Liquidity Risk Management Rule, can provide policy guidance on how the Federal Reserve can prevent Treasury fragility in the future.

Our study is also related to some contemporaneous studies on the economic mechanisms underlying the COVID-19 Treasury market turmoil, particularly on why the Treasury market failed to follow the crisis playbook in March 2020. For example, [Duffie \(2020\)](#) emphasizes the frictions in the market-making mechanism, whereas [Schrimpf, Shin, and Sushko \(2020\)](#) and [Kruttili et al. \(2021\)](#) highlight the role of margin spirals and hedge funds. [He, Nagel, and Song \(2022\)](#) focus on the interaction between leveraged investors financing with repo and dealers subject to balance sheet constraints. We complement this strand of literature by providing a novel perspective from liquidity management of bond funds and the 2017 Liquidity Risk Management Rule. Specifically, we argue that when bond funds, particularly those with large illiquid corporate bond holdings (e.g., [Falato, Goldstein, and Hortaçsu 2021](#)), experienced large fund outflows during the COVID-19 pandemic, liquidity management induced dramatic fire sales on Treasuries from those funds, which at least partially contributed to the Treasury market turmoil during the COVID-19 pandemic. Also, the 2017 Liquidity Risk Management Rule intensified bond funds' liquidity management before the COVID-19 crisis, which in turn strengthened the sales of Treasuries during the crisis. Another important difference between our study and this strand of literature is that we focus on a long sample period and use data with rich information (e.g., detailed bond holdings and fund characteristics). The more generalized setting not only allows us to conduct

cross-sectional tests to pin down the underlying mechanism but also helps demonstrate that liquidity management, together with the fast-growing bond fund sector, has contributed to increased fragility in the Treasury market over the past decades.

A few papers also examine the COVID-19 episode and the corporate bond market rather than the Treasury market (our focus). For example, [Kargar et al. \(2021\)](#) and [O'Hara and Zhou \(2021\)](#) examine the liquidity and micro-structure of the corporate bond market. [Haddad, Moreira, and Muir \(2021\)](#) find that corporate bonds with better credit ratings tended to exhibit more severe price crashes, which were likely driven by the selling pressure from mutual funds. [Jiang et al. \(2022\)](#) find that corporate bonds with higher latent fragility, measured by the asset illiquidity of their mutual fund holders, experienced more negative returns in March 2020.

Finally, our paper is related to the large body of literature on the role of institutional trading in generating price impacts and financial fragility. [Edmans, Goldstein, and Jiang \(2012\)](#) and [Lou \(2012\)](#) show that fund flow-induced trading has a significant price impact on stock markets. [Jiang \(2023\)](#) highlights the usage of high leverage, which can generate crash risk in stock prices. [Anton and Polk \(2014\)](#) show that fund common ownership forecasts the return correlation between stocks. [Greenwood and Thesmar \(2011\)](#) estimate the correlation between fund flows among mutual funds and link the correlated fund flows to the stock return comovement. [Huang, Song, and Xiang \(Forthcoming\)](#) document that the correlation between mutual fund flows contributes to a large portion of the variance-covariance in anomaly returns. Our study contributes to this literature by focusing on the role of liquidity management and bond funds. We find that liquidity management may lead to excess volatility and exacerbate the contagion effect during market turmoil, even in the most liquid market.

1. Background, Data, and Methodology

1.1 Background

Recent decades have witnessed the fast growth of open-end mutual funds that invest in relatively illiquid assets (e.g., corporate bonds). According to a report by the [Investment Company Institute \(2022\)](#), the total assets under management of worldwide regulated open-end bond mutual funds increased from 6.1 trillion USD in 2010 to about 13.7 trillion USD in 2021.

At the same time, the total mutual fund ownership of Treasuries also increased dramatically, making mutual funds an important component in the Treasury market. Within the entire Morningstar bond fund universe, the total ownership of outstanding Treasury securities rose from about 5% in 2002 to more than 13% in 2021. This pattern is also confirmed by [Liang \(2020\)](#), who documents that the number of marketable Treasury shares held by long-term mutual funds in 2019 exceeded the amount held by banks and broker-dealers.

Unlike other major investors in the Treasury market—such as insurance companies, pensions, or sovereign wealth funds, who tend to be passive and to buy and hold—open-end mutual funds trade actively in the Treasury market to accommodate fund flows. Such trade-to-flow sensitivity could be further amplified by liquidity management. Thus, the trading of open-end mutual funds could generate price impacts on Treasuries. This argument echoes several recent studies, such as that of [Brooks, Katz, and Lustig \(2020\)](#).

1.2 Sample construction

We obtain data on global actively managed open-end corporate bond mutual funds (termed “bond funds” hereafter) from Morningstar, which include detailed information on fund characteristics, such as fund returns and total net assets (*TNA*), as well as bond funds’ portfolio holdings from 2002Q3 at the quarterly frequency. To obtain a full list of global bond funds, we start with the entire Morningstar universe and apply the following sample filters: (1) we keep funds with Morningstar Global Board Category Group of “Fixed Income”; (2) we exclude exchange-traded funds (ETFs) and keep only open-end funds (Morningstar universe code equals “FO”); (3) we keep funds focusing on corporate bond investment by dropping funds with Morningstar categories that contain the following keywords: “mortgages,” “backed,” “government,” “money market,” “equity-biased,” “inflation,” “banking,” “bank loan,” “muni,” and “preferred stock”; (4) we drop funds with legal structures that are not considered to be open-end mutual funds (i.e., FCPE, open-ended protected cell company, and unit trusts);¹³ and (5) we keep funds with US Treasury holding histories.

Note that our sample includes passive index funds. The motivation is based on recent work of [Koont et al. \(2023\)](#) and [Choi, Cremers, and Riley \(2024\)](#). Both studies show that passive funds in fixed income markets actively manage their portfolios, trading off index tracking against liquidity transformation. In this sense, index funds also have a liquidity management motive, and thus, we include index bond funds in our sample. We find that index funds and actively managed bond funds in our sample have similar turnover in Treasuries. Also, our main results are robust to excluding index funds (see [Appendix Table A9](#)).¹⁴

We summarize how each sample filter affects the number of funds in our sample in [Appendix Table A2](#), panel A. Our final sample includes 5,667 unique bond funds worldwide from 2002Q3 through 2021Q4 (identified by a Morningstar portfolio identifier, i.e., *masterportfolioid*). Note that

¹³ FCPE stands for *Fonds communs de placement d'entreprise* and refers to investment funds dedicated to the employees of a company in France.

¹⁴ The average quarterly fund turnover in Treasuries for index funds is 0.053, while that for non-index funds is 0.052. Also, index funds make up a small proportion of our sample (142 out of 5,667 bond funds based on Morningstar categorization).

after applying these filters, Treasury bond funds and money market funds are excluded from the sample.

In [Appendix Table A3](#), we provide descriptive statistics on our bond fund sample across different dimensions, including domicile countries (panel A), base currencies (panel B), Morningstar major holding types (panel C), and Morningstar fixed-income style (panel D). In a nutshell, bond funds in our sample mainly come from the United States and other developed countries, are mainly based on USD and EUR, invest mostly in fixed-income assets, are less interest-rate sensitive, and are more exposed to credit risks.

We obtain data on U.S. Treasuries from the Center for Research in Security Prices (CRSP). During our sample period of 2002Q4 through 2021Q4, there were 2,435 Treasuries in CRSP.¹⁵ At each quarter, we require each Treasury security to have (1) at least 30 nonmissing daily returns, (2) nonmissing price and shares outstanding to compute mutual fund ownership, and (3) more than 6 months remaining time-to-maturity. Our final sample contains 1,623 unique U.S. Treasuries (identified by bond CUSIP).¹⁶ The effect of each sample filter on the number of U.S. Treasuries in our sample is summarized in [Appendix Table A2](#), panel B.¹⁷

The exclusion of Treasuries with time-to-maturity of less than 6 months is motivated by practice among practitioners and prior studies. Particularly, well-known fixed income indices, such as J.P. Morgan fixed-income indices, include only bonds with remaining maturity of 6 months or more (e.g., [J.P. Morgan 2020](#)). This choice is unsurprising as bonds near maturity (within 6 months) are rarely traded and lack liquidity. The lack of liquidity makes these bonds behave oddly in yields and more subject to erroneous prices (e.g., [Gebhardt, Hvidkjaer, and Swaminathan 2005](#); [Gürkaynak, Sack, and Wright 2007](#)). As a result, prior studies (e.g., [Harris and Piwowar 2006](#); [Todorov 2020](#); [Huynh and Xia 2021](#); [Bali, Subrahmanyam, and Wen 2021](#); [Ma, Xiao, and Zeng 2022](#)) often exclude bonds maturing within 6 months or 1 year. After carefully considering the choice between 6 months and 1 year, we selected 6 months as the cutoff in our main analysis to keep our sample as close as possible to the entire CRSP sample. Nevertheless, we also consider alternative cutoffs (i.e., 9 months and 1 year) for robustness and find similar results as reported in [Appendix Table A6](#).

1.3 Fund flows, net buy, and ownership

For each fund at each quarter, we calculate fund flows as follows:

$$Fund\ Flow_{f,q} = \frac{TNA_{f,q} - TNA_{f,q-1} \cdot (1 + Fund\ Return_{f,q})}{TNA_{f,q-1}}, \quad (1)$$

¹⁵ We start the sample at 2002Q4 because our fund holding data starts at 2002Q3.

¹⁶ We have confirmed that our sample of Treasuries is representative: Treasuries in our sample have maturities and coupon rates similar to all Treasuries covered by CRSP from 2002Q4 through 2021Q4.

¹⁷ As our primary focus is on Treasuries, for brevity we do not discuss detailed information on sample construction of corporate bonds here, but address it in [Section A](#) of the [Appendix](#).

where *Fund Return*_{*f,q*} is fund *f*'s gross return over quarter *q*, and *TNA*_{*f,q*} is fund *f*'s total net assets at quarter *q*. Morningstar provides returns at the share class level; thus we take value-weighted average returns across all share classes to calculate *Fund Return*_{*f,q*} for each fund.

Next, using bond fund holding data, we calculate how bond funds trade three different asset classes: U.S. Treasuries, other government bonds, and corporate bonds. We identify U.S. Treasuries by matching holding details to the CRSP by CUSIP. For other government bonds, we first identify holdings with the following Morningstar detailed holding type identifiers: Government/Treasury (BT), Government Inflation Protected (TP), Supranational (BZ), Subsovereign Government Debt (DS), and Short-Term Government Bills (GS); then we exclude U.S. Treasuries. In our sample, the majority of these other government bonds are issued by developed countries. For corporate bonds, we include holdings with the following Morningstar detailed holding type identifiers: Corporate bond (B), Corporate Information Protected (IP), Bank Loans (BR), Capital Contingent Debt (CT), Convertibles (BC), Contingent Convertible (CN), and Short-Term Corporate Bills (BB).

For each asset class at each quarter, we follow prior studies (e.g., Lou 2012) and define *Net Buy* as

$$Net\ Buy_{f,q} = \frac{\sum_i^N Share_{i,f,q} - \sum_i^N Share_{i,f,q-1}}{\sum_i^N Share_{i,f,q-1}}, \quad (2)$$

where *Share*_{*i,f,q*} is the amount of bond *i* held by fund *f* at quarter *q*, and *N* is the total number of bonds in the asset class. By construction, *Net Buy*_{*f,q*} measures the percentage change of a fund's total holdings in an asset class relative to its beginning-of-the-quarter holdings. Note that we exclude price in Equation (2), and thus our measure, *Net Buy*_{*f,q*}, is driven purely by funds' trading on each asset class rather than by changes in security prices, teasing out the mechanical portfolio rebalancing effect.¹⁸

We construct fund ownership and common ownership as follows. First, for each bond in each quarter, we calculate bond fund ownership (denoted *Ownership*) as the ratio of the market value held by all bond funds to the total market value outstanding. Then, we follow Anton and Polk (2014) and calculate bond fund common ownership (denoted *Common Ownership*) to measure the extent to which a pair of bonds is held by the same funds (termed "common funds"). Specifically, for a pair of bonds *i* and *j* at each quarter, *Common Ownership* is defined as:

$$Common\ Ownership_{i,j,q} = \frac{\sum_{f=1}^F (Shares_{i,f,q} \cdot P_{i,q} + Shares_{j,f,q} \cdot P_{j,q})}{Shares_{Outstanding_{i,q}} \cdot P_{i,q} + Shares_{Outstanding_{j,q}} \cdot P_{j,q}}, \quad (3)$$

¹⁸ In Appendix Table A4, we take into account bond prices in the formula when calculating *Net Buy*_{*f,q*} and find similar results, suggesting that portfolio rebalance does not affect our results.

where $Shares_{i,f,q}$ is the amount of bond i held by fund f in quarter q , F is the number of funds holding both bonds i and j , $SharesOutstanding_{i,q}$ is the total amount outstanding of bond i at quarter q , and $P_{i,q}$ is the price of bond i at quarter q .

1.4 Liquidity management intensity and fund characteristics

To investigate the mechanism of liquidity management behind the association between bond fund ownership and the excess return volatility of U.S. Treasuries, we estimate a fund-level measure of liquidity management intensity (LMI) from rolling regressions. Specifically, for each fund f in each quarter q , we run the following univariate regression using observations from the past 12 quarters:¹⁹

$$Net\ Buy\ (US\ Treasuries)_{f,q} = \alpha + \beta \cdot Fund\ Flow_{f,q} + \varepsilon_{f,q}. \quad (4)$$

Here, $Net\ Buy\ (US\ Treasuries)_{f,q}$ is fund f 's trading on U.S. Treasuries in quarter q ; $Fund\ Flow_{f,q}$ is fund f 's net flows in quarter q . The regression coefficient, β , captures the sensitivity of fund f 's Treasury trading in response to fund flows; LMI equals the coefficient estimate of β . Intuitively, if funds use Treasuries for buffer and smooth flow shocks, β should be greater than one. For each quarter, we sort all funds on LMI and label the top tercile high-LMI funds, the rest are labeled low-LMI funds.

To investigate the determinants of LMI, we consider the following fund characteristics that could be associated with liquidity management intensity: (1) *Cash Weight*, the percentage of cash-type assets in a fund's portfolio;²⁰ (2) *Fund Flow*, the quarterly net flow of the fund; (3) *Fund Returns*, the quarterly gross return of the fund; (4) *Fund Flow Volatility*, the standard deviation of quarterly fund flows from the past 12 quarters; (5) *US Fund*, a dummy variable that equals one if the domicile of the fund is the United States, and zero otherwise; and (6) *Log(Age)*, the natural logarithm of fund age.

In addition, in a spirit similar to that of Jiang et al. (2022), we construct three proxies for fund illiquidity: *Fund Amihud*, *Fund IRC* (imputed round-trip costs), and *Fund Spread*. For example, *Fund Amihud* is computed as:

$$Fund\ Amihud_{f,q} = \sum_{i=1}^N (Holding\ Weight_{i,q} \cdot Bond\ Amihud_{i,q}), \quad (5)$$

where $Holding\ Weight_{i,q}$ is the portfolio weight of U.S. corporate bond i in quarter q , and $Bond\ Amihud_{i,q}$ is the Amihud (2002) measure for U.S.

¹⁹ We require at least five non-missing observations to run this regression. Because of this requirement, our LMI measures starts from 2004Q2.

²⁰ We identify cash-type assets using the following Morningstar detail holding type identifiers: C (Cash), CD (Cash-CD/Time Deposit), CA (Cash-Collateral), CP (Cash-Commercial Paper), CH (Cash-Currency), OT (Cash-Forward Offset), CQ (Cash-Future Offset), CC (Cash-Option (Call)), CO (Cash-Option (Put)), OO (Cash-Option Offset), CR (Cash-Repurchase Agreement), SV (Cash-Stable Value Fund), and OS (Cash-Swap Offset).

corporate bond i in quarter q . To compute $Bond\ Amihud_{i,q}$, we first remove a trade if its price changed more than 20% from the previous trade within the same day (to avoid potential data errors). Then, we compute the Amihud measure as the absolute value of return divided by the trading volume for each transaction, and then average across all trades of a bond within a quarter. $Fund\ IRC$ and $Fund\ Spread$ are constructed similarly, by replacing $Bond\ Amihud_{i,q}$ with $Bond\ IRC_{i,q}$ and $Bond\ Spread_{i,q}$, respectively. $Bond\ IRC_{i,q}$ refers to the quarterly imputed round-trip costs calculated for each U.S. corporate bond following Dick-Nielsen, Feldhütter, and Lando (2012). $Bond\ Spread_{i,q}$ is the same-bond-same-day effective bid-ask spread calculated following Hong and Wargha (2000), which equals the average buy prices minus the average sell prices of all transactions on the same day for the same bond. We first calculate the measure for each bond for each day and then the average for each bond for all days within a quarter. Overall, these three measures capture both the liquidity conditions of the corporate bonds and their weights in a fund's portfolio.

1.5 Risk-adjusted returns, volatility, and correlation

We measure the excess return volatility and excess return comovement of U.S. Treasuries using daily risk-adjusted returns. First, we calculate daily bond returns by adjusting price changes with accrued interests (AI) and coupon payments (C). More precisely, the daily return for bond i at day d is given by:

$$Bond\ Return_{i,d} = \frac{P_{i,d} + AI_{i,d} + C_{i,d}}{P_{i,d-1} + AI_{i,d-1}} - 1. \quad (6)$$

Here, $P_{i,d}$ is the clean price (or the average of bid and ask, if the clean price is missing) at the day-end from the CRSP.

Second, for each bond, we compute daily risk-adjusted returns as the residuals from either a market model (following Frazzini and Pedersen 2014) or a three-factor model:

$$Bond\ RetRF_{i,d} = \alpha_i + \sum_{s=0}^2 \beta_{i,d-s} TRY_{d-s} + \varepsilon_{i,d}, \quad (7)$$

$$Bond\ RetRF_{i,d} = \alpha_i + \sum_{s=0}^2 \beta_{i,d-s} TRY_{d-s} + \sum_{s=0}^2 \gamma_{i,d-s} IG_{d-s} + \sum_{s=0}^2 \theta_{i,d-s} HY_{d-s} + \varepsilon_{i,d}, \quad (8)$$

where $Bond\ RetRF_{i,d}$ is bond i 's daily return minus the risk-free rate on day d , and the risk-free rate is the daily rate of the 1 month Treasury bill. On the right-hand side of Equation (7), we follow the well-known study of Frazzini and Pedersen (2014) and use one return factor, the aggregate daily

returns of Treasuries (in excess of the risk-free rate). On the right-hand side of Equation (8), we use as return factors the aggregate daily returns of three major bond categories: Treasuries, investment-grade bonds, and junk bonds. This choice is motivated as follows. First, since our main variable of interest is return volatility (rather than yield volatility), the ideal risk factors should be based on Treasury returns. Thus, we follow prior studies (e.g., Frazzini and Pedersen 2014) to consider an equally weighted Treasury bond excess market return as a benchmark return. Second, since our study also attempts to show how corporate bond funds exert non-fundamental fragility on Treasuries, we would like to control for the tradable benchmark returns of corporate bonds, and thus we follow Choi, Kronlund, and Oh (2022) and include investment-grade and high-yield bond market returns as benchmark returns. Specifically, we use returns from the Bloomberg U.S. Corporate Bond Index (Ticker: LUACTRUU) to proxy for the aggregate returns from investment-grade bonds and use returns from the Bloomberg U.S. Corporate High Yield Bond Index (Ticker: LF98TRUU) to proxy for the aggregate returns from junk bonds. We denote these three factors as *TRY*, *IG*, and *HY*, respectively. In addition, we include two lags for each factor to take into account non-synchronized trading.²¹ As a robustness test, we add three term-structure of government bond yields and their two lags in Equation (8), following Chen, Ferson, and Peters (2010). The three term-structure of government bond yields include a short-term interest rate (proxied by the 3 month Treasury rate), a measure of the term slope (proxied by the difference between the 10 year yield and the 1 year yield), and a measure of the curvature of the yield curve (proxied by $y_3 - (y_7 + 2y_1)/3$, where y_i is the i -year fixed-maturity yield). Data to compute these term structure of government bond yields is obtained from the Federal Reserve H.15 release.

Third, with the daily risk-adjusted returns, we compute the standard deviation of daily risk-adjusted returns (either market-adjusted or three-factor-adjusted) in a quarter for each U.S. Treasury, denoted as *Volatility*, as our key proxy for Treasury fragility. To capture the excess return comovement, we calculate the pairwise correlation between a pair of U.S. Treasuries using daily risk-adjusted returns, denoted as *Corr_all*. To examine the asymmetry in the excess return comovement, within each quarter, we split all trading days into two groups (downside markets and upside markets) based on the aggregate Treasury market returns (i.e., *TRY*). We then calculate the excess return comovement among U.S. Treasuries using daily risk-adjusted returns in each group, denoted as *Corr_down* and *Corr_up*. Finally, we take the difference in the excess return comovement between downside and upside markets, denoted as *Down-minus-up*. These pairwise correlation measures are also calculated for corporate bond pairs for comparison.

²¹ Within each quarter, we drop the bonds with fewer than 30 nonzero trading days in the quarter.

1.6 Summary statistics

Table 1 reports summary statistics. As shown in panel A, the market size of bond funds has expanded quickly over time. The number of bond funds increased from 1,070 in 2002 to 3,436 in 2021. The total AUM of all bond funds multiplied nearly 10 times, from 637 billion USD in 2002 to more than 6.3 trillion USD in 2021. Panel B reports summary statistics for bond funds that are included in the trade-to-flow analysis. The average quarterly fund flow is about 1% and the average quarterly fund return is about 0.6%. Panel C

Table 1
Summary statistics

A: Time trend for bond funds						
Year	# of bond funds	Average fund TNA (\$M)	Median fund TNA (\$M)	Total AUM (\$B)		
2002	1,070	595	148	637		
2003	1,308	609	151	797		
2004	1,428	632	160	903		
2005	1,936	633	159	1,128		
2006	2,073	644	139	1,335		
2007	2,302	672	147	1,547		
2008	2,429	565	115	1,371		
2009	2,453	792	153	1,944		
2010	2,507	973	168	2,440		
2011	2,622	1,016	163	2,663		
2012	2,733	1,241	193	3,392		
2013	2,902	1,274	185	3,698		
2014	3,076	1,186	191	3,647		
2015	3,236	1,112	170	3,598		
2016	3,353	1,167	175	3,914		
2017	3,440	1,362	203	4,685		
2018	3,512	1,289	189	4,529		
2019	3,527	1,522	217	5,367		
2020	3,475	1,771	260	6,156		
2021	3,436	1,842	264	6,328		
B: Summary statistics for bond funds						
Variable	Mean	Std	P25	P50	P75	N
Net Buy (US Treasuries)	0.065	0.680	−0.231	0.000	0.195	72,528
Net Buy (Other Government Bonds)	0.104	0.581	−0.110	0.000	0.138	72,528
Net Buy (Corporate Bonds)	0.041	0.320	−0.071	0.003	0.093	72,528
Fund Flow	0.010	0.116	−0.054	−0.002	0.056	72,528
Fund Return	0.006	0.027	−0.004	0.007	0.020	72,528
C: Summary statistics for U.S. Treasuries						
Variable	Mean	Std	P25	P50	P75	N
Volatility (in %, market-adjusted)	0.065	0.066	0.027	0.046	0.070	17,720
Volatility (in %, three-factor-adjusted)	0.059	0.060	0.025	0.042	0.064	17,720
Ownership	0.046	0.041	0.019	0.037	0.061	17,720
Time-to-maturity	6.716	7.496	1.833	3.877	7.632	17,720
Coupon Rate	3.300	2.509	1.500	2.625	4.500	17,720
On-the-run	0.075	0.263	0.000	0.000	0.000	17,720
Log(Size)	10.209	0.569	9.962	10.307	10.531	17,720
Bid-ask Spread	0.042	0.017	0.031	0.039	0.047	17,720

(continued)

Table 1
Continued

D: Summary statistics for U.S. Treasury pairs						
Variable	Mean	Std	P25	P50	P75	N
<i>Corr_all</i> (market-adjusted)	0.153	0.551	−0.231	0.234	0.625	2,040,889
<i>Corr_up</i> (market-adjusted)	0.152	0.558	−0.252	0.238	0.630	2,040,889
<i>Corr_down</i> (market-adjusted)	0.152	0.557	−0.252	0.241	0.630	2,040,889
<i>Down-minus-up</i> (market-adjusted)	0.000	0.202	−0.112	−0.001	0.113	2,040,889
<i>Corr_all</i> (three-factor-adjusted)	0.149	0.544	−0.231	0.228	0.611	2,040,889
<i>Corr_up</i> (three-factor-adjusted)	0.147	0.550	−0.255	0.232	0.616	2,040,889
<i>Corr_down</i> (three-factor-adjusted)	0.149	0.551	−0.249	0.234	0.620	2,040,889
<i>Down-minus-up</i> (three-factor-adjusted)	0.002	0.200	−0.108	0.002	0.114	2,040,889
<i>Common Ownership</i>	0.017	0.014	0.004	0.015	0.026	2,040,889
<i>Time-to-maturity Difference</i>	7.160	7.915	1.455	3.586	10.260	2,040,889
<i>Coupon Rate Difference</i>	2.156	2.040	0.625	1.500	3.125	2,040,889
<i>On-the-run Difference</i>	0.161	0.367	0.000	0.000	0.000	2,040,889
<i>Log(Size) Difference</i>	0.507	0.458	0.156	0.372	0.756	2,040,889
<i>Bid-ask Spread Difference</i>	0.017	0.016	0.000	0.016	0.031	2,040,889

This table reports descriptive statistics. Panel A reports the time trend of the bond fund sector, including the number of bond funds, average (median) fund size, and the total assets under management for all bond funds in our sample. Panel B reports the summary statistics for the fund-level trade-to-flow analyses. *NetBuy (US Treasuries)*, *NetBuy (Other Government Bonds)*, and *NetBuy (Corporate Bonds)* measure the percentage change of a fund's total holdings in U.S. Treasuries, other government bonds, and corporate bonds, relative to its beginning of the quarter holdings, respectively. *Fund Flow* is the quarterly fund flows. *Fund Return* is the quarterly fund return. Panel C reports the summary statistics for U.S. Treasuries. *Volatility (in %)* is the standard deviation (in percentage) of the daily risk-adjusted returns in a quarter. We consider two ways to compute risk-adjusted returns: market-adjusted and three-factor-adjusted. Market-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market and its two lags; three-factor-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market, investment-grade bonds, junk bonds, and their two lags. *Ownership* is the proportion of total market value of a Treasury that is held by bond funds. *Time-to-maturity* is the years between the quarter-end and maturity date. *Coupon Rate* is the coupon rate expressed as a percentage. *On-the-run* is a dummy variable that equals one if a Treasury is the most recently issued Treasury of a particular maturity, and zero otherwise. *Log(Size)* is the logarithm of the total amount outstanding (face value, in millions of USD). *Bid-ask Spread* is the difference between ask price and bid price. Panel D reports the summary statistics for Treasury pairs. *Corr_all* is the excess return correlation between two Treasuries in a quarter. The excess return correlation is computed as the pairwise correlation of daily risk-adjusted returns for a pair of Treasuries. We further sort all trading days in a quarter into two equal groups (downside and upside market days) based on the aggregate Treasury market returns, and *Corr_up* and *Corr_down* are the excess return correlation between two Treasuries during upside and downside market days, respectively. *Down-minus-up* is the difference between *Corr_down* and *Corr_up*. All these pairwise correlations are constructed using market-adjusted returns and three-factor-adjusted returns. *Common Ownership* is the proportion of total market value of a Treasury pair held by funds holding the pair of Treasuries. *Time-to-maturity Difference* is the absolute difference between two Treasuries' years-to-maturity. *Coupon Rate Difference* is the absolute difference between two Treasuries' coupon rates. *On-the-run Difference* is the absolute difference between two Treasuries' on-the-run status for a pair of Treasuries. *Log(Size) Difference* is the absolute difference between two Treasuries' logarithm of total amount outstanding. *Bid-ask Spread Difference* is the absolute difference between two Treasuries' bid-ask spread. The sample period is from 2002Q4 through 2021Q4.

reports summary statistics on U.S. Treasuries. The average volatility of market-adjusted daily returns is about 0.065% (or 1.03% annualized), and the average quarterly bond fund ownership is 4.6%. Panel D reports summary statistics of the variables for Treasury pairs in the excess return comovement analyses. The average market-adjusted excess return correlation is 15.3% for pairs of Treasuries.

2. Main Results

2.1 Liquidity management with U.S. Treasuries

We first examine the liquidity management of bond funds. Our analysis is similar to prior studies on liquidity management, for example, [Chernenko and Sunderam \(2016\)](#), [Choi et al. \(2020\)](#), and [Jiang, Li, and Wang \(2021\)](#). While those studies examine cash and cash-like securities, we focus on the role of U.S. Treasuries in liquidity management. As we will show later, using U.S. Treasuries as a tool of liquidity management has important and testable asset pricing implications.

Like banks, bond funds perform liquidity transformation and are subject to potential investor run. That is, while bond funds invest heavily in illiquid assets (e.g., corporate bonds), they issue liquid claims (fund shares) that investors can redeem at the net asset value (*NAV*) on a daily basis. This liquidity mismatch between fund shares and the underlying assets can generate strategic complementarities among fund investors, leading to financial fragility of funds (e.g., [Chen, Goldstein, and Jiang 2010](#); [Goldstein, Jiang, and Ng 2017](#)). To mitigate this fragility, bond funds actively manage their liquidity (see [Chernenko and Sunderam 2016](#); [Choi et al. 2020](#); [Jiang, Li, and Wang 2021](#)). We argue that U.S. Treasuries play an important role in liquidity management because they are the most liquid assets and trading them has a limited effect on price. In fact, liquidity management is common not only among bond funds but also among other open-end funds holding illiquid assets (e.g., bank loan funds and real estate funds) and even among commercial banks (e.g., [Chen et al. Forthcoming](#); [Ma, Xiao, and Zeng 2022](#)).

In addition to Treasuries, it is also worth examining flow-induced trading of other government bonds. In our sample, the other government bonds are issued mostly by developed countries and have better liquidity than corporate bonds, but are not as liquid as U.S. Treasuries. Based on our hypothesis and on liquidity management practices, bond funds may also prioritize trading other government bonds to accommodate fund flows, but to a lesser extent than in the case of U.S. Treasuries. In this sense, the position of other government bonds can serve as the middle layer between Treasuries and corporate bonds to buffer liquidity shocks.

To verify and quantify the role of U.S. Treasuries in liquidity management, we examine how bond funds trade U.S. Treasuries, other government bonds, and corporate bonds in response to fund flows. We illustrate our test design with the following simplified example. Suppose that a fund has total net assets of \$100 at the beginning of the quarter and it allocates \$20 to U.S. Treasuries, \$10 to other government bonds, and \$70 to corporate bonds. Then, suppose there is a 10% outflow during the quarter. If the fund manager does not engage in liquidity management, they will proportionally liquidate the holdings in all three asset classes. That is, the fund will sell \$2 of U.S. Treasuries, \$1 of other government bonds, and \$7 of corporate bonds. As a result, the positions in all

three asset classes will decrease by 10%, where the trade-to-flow sensitivity is one on U.S. Treasuries, other government bonds, and corporate bonds.

In contrast, if the fund wants to avoid large price impacts in trading corporate bonds, it will prioritize selling government bonds, especially U.S. Treasuries. That is, the fund is likely to liquidate relatively more Treasuries and other government bonds than corporate bonds, for example, selling \$7 of U.S. Treasuries, \$2 of other government bonds, and \$1 of corporate bonds. As a result, the fund's total holding of U.S. Treasuries and other government bonds will decrease by more than 10%, while that of corporate bonds will decrease by less than 10%. In this case (with liquidity management), the trade-to-flow sensitivity is greater than one on U.S. Treasuries and other government bonds but is less than one on corporate bonds.

Motivated by this example, we conduct the following regression to formally examine how bond funds trade U.S. Treasuries in response to fund flows:

$$\begin{aligned} \text{Net Buy (US Treasuries)}_{f,q} = & \alpha + \beta_1 \cdot \text{Fund Flow}_{f,q} + \beta_2 \cdot \text{Fund Flow}_{f,q-1} \\ & + \gamma_1 \cdot \text{Fund Return}_{f,q} \\ & + \gamma_2 \cdot \text{Fund Return}_{f,q-1} + \delta_q + \varepsilon_{f,q}, \end{aligned} \quad (9)$$

where $\text{Net Buy (US Treasuries)}_{f,q}$ is fund f 's trading on U.S. Treasuries in quarter q , $\text{Fund Flow}_{f,q}$ is fund f 's net flows in quarter q , and $\text{Fund Return}_{f,q}$ is the gross return of the fund f in quarter q . We include year-quarter fixed effects and cluster standard errors by funds. The trade-to-flow sensitivity of U.S. Treasuries is indicated by β_1 . We also conduct the same analyses for the other two assets classes: other government bonds and corporate bonds. That is, we replace the dependent variable by $\text{Net Buy (Other Government Bonds)}_{f,q}$ or $\text{Net Buy (Corporate Bonds)}_{f,q}$. As illustrated in the aforementioned example, if funds use Treasuries as the liquidity buffer, β_1 should be greater than one for both U.S. Treasuries and other government bonds but less than one for corporate bonds. Moreover, we should observe a β_1 that is greater for U.S. Treasuries than for other government bonds, because U.S. Treasuries have better liquidity. To make these comparisons on β_1 , we require fund quarters to have nonmissing net trading on all three asset classes.

Table 2 reports the results. Columns (1)–(2) are for U.S. Treasuries, columns (3)–(4) are for other government bonds, and columns (5)–(6) are for corporate bonds. We find that the trade-to-flow sensitivity is greater than one for both U.S. Treasuries and for other government bonds. More importantly, bond funds' trading of U.S. Treasuries is more sensitive to fund flows than it is for other government bonds. For example, as shown in columns (1) and (3), a 1% fund inflow is associated with a 1.42% increase in U.S. Treasury holdings (t -statistic = 48.6) and a 1.18% increase in the holding of other government bonds (t -statistic = 47.7). In contrast, for corporate bonds, the trade-to-flow sensitivity is less than one. As shown in column (5), a 1% fund

Table 2
Fund trading and fund flows

DepVar:	Net Buy (US Treasuries) _{f,q}		Net Buy (Other Government Bonds) _{f,q}		Net Buy (Corporate Bonds) _{f,q}	
	(1)	(2)	(3)	(4)	(5)	(6)
Fund Flow _{f,q}	1.420*** (48.6)	1.458*** (45.5)	1.175*** (47.7)	1.136*** (43.9)	0.843*** (57.2)	0.780*** (51.8)
Fund Flow _{f,q-1}		-0.156*** (-5.9)		0.130*** (6.3)		0.217*** (16.0)
Fund Return _{f,q}	-0.051 (-0.3)	-0.097 (-0.5)	0.677*** (5.8)	0.676*** (5.8)	0.151** (2.2)	0.159** (2.4)
Fund Return _{f,q-1}		0.661*** (4.0)		0.133 (1.2)		0.066 (0.9)
Year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs	72,528	72,528	72,528	72,528	72,528	72,528
Adj R ²	0.080	0.081	0.090	0.091	0.130	0.136

This table reports the regression results of fund trading on fund flows for U.S. Treasuries, other government bonds, and corporate bonds. *NetBuy (US Treasuries)*, *NetBuy (Other Government Bonds)*, and *NetBuy (Corporate Bonds)* measure the percentage share change of a fund's total holdings in U.S. Treasuries, other government bonds, and corporate bonds, relative to its beginning of the quarter holdings, respectively. *Fund Flow* is the quarterly fund flows. *Fund Return* is the quarterly fund return. All variables are winsorized by quarter at the 5th and 95th percentiles. We control for year-quarter fixed effects and standard errors are clustered by funds. *t*-statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. The sample period is from 2002Q4 through 2021Q4.

inflow is associated with an increase in corporate bond holdings of only 0.84% (t -statistic = 57.2). Overall, results presented in Table 2 demonstrate that bond funds prioritize the trading of U.S. Treasuries to actively manage their liquidity conditions. More importantly, the pecking order—as suggested from trade-to-flow sensitivities on U.S. Treasuries, other government bonds, and corporate bonds—is consistent with liquidity management and assures us that the high trade-to-flow sensitivity on Treasuries is not a coincidence. In Appendix Table A4, we construct *Net Buy_{f,q}* as the proportional change in the market values of bonds held by a fund on a quarterly basis, and we find similar results.

2.2 Bond fund ownership and excess return volatility

Having confirmed that bond funds use U.S. Treasuries in liquidity management, we study the asset-pricing implications of such liquidity management in this subsection. As bond funds trade U.S. Treasuries in response to fund flows, they will transmit nonfundamental demand shocks from fund flows into the prices of Treasuries. Thus, the prices of Treasuries held heavily by bond funds should experience excessive volatility. As mutual funds have become major active traders in the Treasury market, such exposure becomes an important driving force for Treasury prices. In our empirical tests, we examine the cross-sectional association between bond fund ownership and U.S. Treasuries' return volatility, which is viewed as a conventional measure of fragility. Also, since we focus on the cross-sectional studies, we can avoid confounding effects in time-series tests and shed some light on the increasing Treasury fragility.

Specifically, we run the following [Fama and MacBeth \(1973\)](#) regressions to examine the effect of bond fund ownership on the return volatility of U.S. Treasuries:

$$Volatility_{i,q} = \alpha + \beta \cdot Ownership_{i,q-1} + \theta \cdot X_{i,q-1} + \varepsilon_{i,q}, \quad (10)$$

where $Volatility_{i,q}$ is the excess return volatility for Treasury i in quarter q (in percentage, using either market-adjusted or three-factor-adjusted daily returns), and the key independent variable is $Ownership_{i,q-1}$, which measures the extent to which bond i is held by bond funds at the end of the previous quarter. We control for *Time-to-maturity*, the years between the quarter-end and maturity date, in all regression specifications to control for the mechanical relation between bond maturity and return volatility. We also control for the coupon rate of the Treasury (denoted as *Coupon Rate*); *On-the-run*, a dummy variable that equals one if the Treasury is the most recently issued Treasury in its maturity class, and zero otherwise; *Log(Size)*, the logarithm of the total amount outstanding (face value, in millions of USD); and *Bid-ask Spread*, the difference between ask and bid prices. We compute [Newey and West \(1987\)](#) standard errors corrected by serial dependence of three lags.

Table 3 reports the results. We find that bond fund ownership is indeed positively associated with subsequent excess return volatility for U.S. Treasuries. For example, column (3) shows that, after controlling for all Treasury characteristics, the coefficient on *Ownership* is 0.070 (t -statistic = 3.7). Such an effect is both statistically significant and economically meaningful: a one-standard-deviation increase in bond fund ownership is associated with a 0.0029% higher volatility of market-adjusted daily returns. For reference, during our sample period from 2002 through 2021, the average market-adjusted excess return volatility of Treasuries was 0.065%. So the effect is about 4.4% relative to our sample mean. Such an effect implies a sizable economic magnitude, which is evident from the comparison between the Treasury and stock markets. [Greenwood and Thesmar \(2011\)](#) find that an increase in mutual fund ownership of 10% leads to an increase in daily stock volatility of about 10% of the sample mean. Column (3) in Table 3 suggests that an increase in mutual fund ownership of 10% is associated with a rise in Treasury return volatility of 10.8% ($= 0.070 \times 10\% / 0.065$) of the sample mean. In this sense, the effect of mutual fund ownership on Treasury return volatility is comparable to, if not greater than, the effect of mutual fund ownership on stock return volatility.²²

Finally, we have explored the cross-sectional heterogeneity in liquidity at the Treasury level. We measure Treasury liquidity using the average daily bid-ask spread in each quarter. As shown in [Appendix Table A5](#), the positive

²² It would be interesting to examine whether bond fund ownership is also associated with the volatility of other government bonds. However, we do not have the price data that allows us to examine the price impact of liquidity management on other government bonds.

Table 3
Bond fund ownership and U.S. Treasury return volatility

DepVar:	Volatility (in %)					
	Market-adjusted returns			Three-factor-adjusted returns		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Ownership</i>	0.053** (2.2)	0.080*** (3.5)	0.070*** (3.7)	0.058** (2.4)	0.082*** (3.6)	0.071*** (3.8)
<i>Time-to-maturity</i>	0.007*** (11.0)	0.007*** (10.9)	0.008*** (12.2)	0.006*** (11.1)	0.006*** (10.9)	0.007*** (12.2)
<i>Coupon Rate</i>		-0.002*** (-4.8)	-0.002*** (-3.1)		-0.002*** (-4.5)	-0.001*** (-2.8)
<i>On-the-run</i>		-0.002** (-2.4)	-0.004*** (-4.2)		-0.002** (-2.2)	-0.003*** (-4.2)
<i>Log(Size)</i>			0.001 (1.0)			0.001 (1.3)
<i>Bid-ask Spread</i>			-0.345*** (-5.1)			-0.314*** (-5.0)
# of Obs	17,720	17,720	17,720	17,720	17,720	17,720

This table reports the results from Fama-MacBeth regressions of U.S. Treasury excess return volatility in quarter q on its bond fund ownership in quarter $q-1$. *Volatility (in %)* is the standard deviation (in percentage) of the daily risk-adjusted returns in a quarter. We consider two ways to compute risk-adjusted returns: market-adjusted and three-factor-adjusted. Market-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market and its two lags; three-factor-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market, investment-grade bonds, junk bonds, and their two lags. *Ownership* is the proportion of total market value of a Treasury that is held by bond funds. *Time-to-maturity* is the years between the quarter-end and maturity date. *Coupon Rate* is the coupon rate expressed as a percentage. *On-the-run* is a dummy variable that equals one if a Treasury is the most recently issued Treasury of a particular maturity, and zero otherwise. *Log(Size)* is the logarithm of the total amount outstanding (face value, in millions of USD). *Bid-ask Spread* is the difference between ask price and bid price. All independent variables (except for the *On-the-run* dummy) are winsorized at the 1st and 99th percentiles. Heteroscedasticity and auto-correlation-consistent Newey-West t -statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. The sample period is from 2002Q4 through 2021Q4.

association between bond fund ownership and Treasury return volatility is most pronounced among Treasuries with mild liquidity conditions (columns (2) and (5)). For the subsample of Treasuries with the worst liquidity conditions (columns (3) and (6)), the coefficient on fund ownership is large in magnitude but statistically insignificant. For the most liquid Treasury group (columns (1) and (4)), the coefficient is of a small magnitude. These results are not surprising and are consistent with the intuition: bond funds tend to use liquid Treasuries for liquidity management, but bond fund trading exerts large (small) price impacts on Treasuries with low (high) liquidity.

2.3 Liquidity management as the mechanism

To investigate the liquidity management mechanism behind the positive association between bond fund ownership and Treasuries' volatility, we link bond funds' liquidity management directly to Treasuries' excess return volatility. Specifically, we take the following steps. First, we estimate a fund-quarter-level proxy of liquidity management intensity as described in Section 1.4. Second, we decompose bond fund ownership into two parts—ownership from the high-LMI funds versus low-LMI funds—and rerun our analyses in Table 3. If the aforementioned findings in Table 3 indeed arose

from liquidity management, our baseline results should come mainly from the ownership of high-LMI funds.

To set the stage, for each fund f in each quarter q , we run a univariate regression of *Net Buy (US Treasuries)* $_{f,q}$ on *Fund Flow* $_{f,q}$ using the samples from the past 12 quarters (at least five nonmissing observations required). The regression coefficient on *Fund Flow* $_{f,q}$ is denoted $LMI_{f,q}$, and it captures how aggressively fund f trades Treasuries in response to fund flows. Intuitively, the more a bond fund conducts liquidity management with Treasuries, the greater its LMI.

We acknowledge that our LMI measure is a statistic of various dimensions of liquidity management motives. Inspired by the literature on mutual fund liquidity management, we conduct Fama-MacBeth regressions to examine the determinants of LMI. We first consider the following fund characteristics that might be correlated with liquidity management intensity: (1) *Cash Weight*, (2) *Fund Flow*, (3) *Fund Returns*, (4) *Fund Flow Volatility*, (5) *US Fund* dummy, and (6) *Log(Age)*. In addition, inspired by Jiang et al. (2022), we also construct three proxies for fund illiquidity: *Fund Amihud*, *Fund IRC*, and *Fund Spread*.²³ Table 4 shows that funds with low cash holding, high flow volatility, and high fund illiquidity tend to exhibit strong liquidity management intensity, consistent with results reported in prior studies (e.g., Chernenko and Sunderam 2016; Choi et al. 2020; Jiang, Li, and Wang 2021).

After justifying the LMI measure, we use it to examine the mechanism behind the positive association between bond fund ownership and Treasury excess return volatility. Specifically, at the end of each quarter, we classify bond funds with LMI in the top tercile as high-LMI funds and the rest as low-LMI funds. Then, for each U.S. Treasury, we calculate its fund ownership separately for high- and low-LMI bond funds, denoted as *Ownership (High LMI)* and *Ownership (Low LMI)*, respectively. Finally, we conduct the same Fama-MacBeth regressions as the ones reported in Table 3 and replace *Ownership* with *Ownership (High LMI)* and *Ownership (Low LMI)*.

Table 5 reports the results and confirms our conjecture. Across all specifications, both high- and low-LMI ownership measures are positively and significantly associated with U.S. Treasuries' excess return volatility. More importantly, the effect is significantly stronger for ownership from high-LMI funds. For example, column (3) shows that the coefficient on *Ownership (High LMI)* is 0.182 (t -statistic = 2.9) and the coefficient on *Ownership (Low LMI)* is only 0.042 (t -statistic = 2.9). The difference, 0.140, is statistically significant at 1% (t -statistic = 3.5). In other words, the effect of ownership on volatility for high-LMI funds is more than four times that for low-intensity funds. These results provide direct evidence supporting the liquidity management mechanism.

²³ For the LMI analyses, the sample period starts from 2004Q2 and ends at 2021Q4, as we require at least five nonmissing observations to estimate the LMI.

Table 4
The determinants of liquidity management intensity

DepVar:	Liquidity Management Intensity (LMI)					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Cash Weight</i>	−0.013*** (−4.7)	−0.014*** (−5.0)	−0.013*** (−4.8)	−0.013*** (−4.5)	−0.013*** (−4.5)	−0.013*** (−4.5)
<i>Fund Flow</i>	0.152 (1.4)	0.038 (0.4)	0.010 (0.1)	0.062 (0.5)	0.049 (0.4)	0.057 (0.5)
<i>Fund Return</i>	−2.272** (−2.4)	−2.294** (−2.4)	−2.735*** (−2.8)	−4.516*** (−3.7)	−4.793*** (−3.9)	−4.853*** (−4.0)
<i>Fund Flow Volatility</i>		0.603*** (11.6)	0.600*** (8.9)	0.586*** (7.3)	0.572*** (7.0)	0.587*** (7.1)
<i>US Fund</i>			0.100*** (3.0)	0.017 (0.6)	0.024 (0.8)	0.014 (0.4)
<i>Log(Age)</i>			−0.069 (−1.3)	−0.095 (−1.4)	−0.095 (−1.4)	−0.093 (−1.4)
<i>Fund Amihud</i>				0.274*** (7.3)		
<i>Fund IRC</i>					0.032*** (7.6)	
<i>Fund Spread</i>						0.029*** (8.1)
# of Obs	73,693	73,693	73,693	59,177	59,145	59,136

This table reports the results from Fama-MacBeth regressions of bond fund liquidity management intensity on its own characteristics. *Liquidity Management Intensity* is the coefficient estimate from a univariate regression of *NetBuy (US Treasuries)* on *Fund Flow* using data from the past 12 quarters, where *NetBuy (US Treasuries)* is the percentage share change of a fund's total holdings in U.S. Treasuries relative to its beginning of the quarter holdings, and *Fund Flow* is the quarterly fund flows. We require at least five quarterly observations to estimate this coefficient. *Cash Weight* is the proportion of cash-type assets in a fund's quarterly holding. *Fund Return* is the quarterly fund return. *Fund Flow Volatility* is the standard deviation of quarterly fund flows from the past 12 quarters. *US Fund* is a dummy variable that equals one if the domicile country of the fund is the United States, and zero otherwise. *Log(Age)* is the natural logarithm of the fund age. *Fund Amihud*, *Fund IRC*, and *Fund Spread* are the holding-weighted sum of Amihud ratio, imputed round-trip transaction cost, and the same-bond-same-day effective spread, respectively, for all U.S. corporate bonds in the quarterly holding. All variables are winsorized by quarter at the 5th and 95th percentiles. Heteroscedasticity and auto-correlation-consistent Newey-West *t*-statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. The sample period is from 2004Q2 through 2021Q4.

We further check the robustness of results from Tables 3 and 5 across various alternative empirical specifications. First, we consider an alternative way to estimate daily excess returns for U.S. Treasuries. In particular, Treasury returns could be affected by term-structure of government bond yields. To address this issue, we include three additional term-structure variables of government bond yields and their respective two lags in Equation (8) when estimating daily excess returns for each Treasury in each quarter. These variables are based on Chen, Ferson, and Peters (2010) and include a short-term interest rate (proxied by the 3-month Treasury rate), a measure of the term slope (proxied by the difference between the 10-year yield and the 1-year yield), and a measure of the curvature of the yield curve (proxied by $y_3 - (y_7 + 2y_1)/3$, where y_i is the i -year fixed-maturity yield). Column (1) of Appendix Table A6 shows that our results are highly similar to our main results.

Second, we consider different sample filters based on the time-to-maturity. In our baseline analyses, we exclude Treasuries with a time-to-maturity of less than 6 months because Treasuries close to maturity are usually illiquid.

Table 5
Decompose ownership by liquidity management intensity and U.S. Treasury return volatility

DepVar:	Volatility (in %)					
	Market-adjusted returns			Three-factor-adjusted returns		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Ownership (High LMI)</i>	0.217** (2.3)	0.175*** (2.9)	0.182*** (2.9)	0.216** (2.4)	0.175*** (2.9)	0.179*** (2.8)
<i>Ownership (Low LMI)</i>	0.006 (0.3)	0.048*** (3.0)	0.042*** (2.9)	0.013 (0.8)	0.050*** (3.3)	0.043*** (3.2)
<i>Time-to-maturity</i>	0.007*** (11.3)	0.007*** (11.2)	0.008*** (12.1)	0.007*** (11.4)	0.007*** (11.2)	0.007*** (12.1)
<i>Coupon Rate</i>		−0.002*** (−5.5)	−0.002*** (−4.1)		−0.002*** (−5.1)	−0.002*** (−3.7)
<i>On-the-run</i>		−0.003*** (−3.2)	−0.004*** (−4.0)		−0.003*** (−3.1)	−0.003*** (−4.0)
<i>Log(Size)</i>			0.000 (0.1)			0.001 (0.5)
<i>Bid-ask Spread</i>			−0.297*** (−5.1)			−0.270*** (−4.9)
# of Obs	16,931	16,931	16,931	16,931	16,931	16,931

This table reports the results from Fama-MacBeth regressions of U.S. Treasury excess return volatility in quarter q on its bond fund ownership in quarter $q - 1$. *Volatility (in %)* is the standard deviation (in percentage) of the daily risk-adjusted returns in a quarter. We consider two ways to compute risk-adjusted returns: market-adjusted and three-factor-adjusted. Market-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market and its two lags; three-factor-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market, investment-grade bonds, junk bonds, and their two lags. We decompose bond fund ownership into two components: *Ownership (High LMI)* is the proportion of total market value of a Treasury that is held by funds with high liquidity management intensity; *Ownership (Low LMI)* is the proportion of total market value of a Treasury that is held by the rest of the funds. Liquidity management intensity is defined as the coefficient estimate from a univariate regression of *NetBuy (US Treasuries)* on *Fund Flow* using data from the past 12 quarters, where *NetBuy (US Treasuries)* is the percentage change of a fund's total holdings in U.S. Treasuries relative to its beginning of the quarter holdings, and *Fund Flow* is the quarterly fund flows. We require at least five quarterly observations to estimate this coefficient. The top one-third of funds in each quarter are identified as funds with high liquidity management intensity based on this estimate. *Time-to-maturity* is the years between the quarter-end and maturity date. *Coupon Rate* is the coupon rate expressed as a percentage. *On-the-run* is a dummy variable that equals one if a Treasury is the most recently issued Treasury of a particular maturity, and zero otherwise. *Log(Size)* is the logarithm of the total amount outstanding (face value, in millions of USD). *Bid-ask Spread* is the difference between ask price and bid price. All independent variables (except for the *On-the-run* dummy) are winsorized at the 1st and 99th percentiles. Heteroscedasticity and auto-correlation-consistent Newey-West t -statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. The sample period is from 2004Q2 through 2021Q4.

In columns (2)–(5) of [Appendix Table A6](#), we consider two alternative maturity filters: 9 months and 1 year. Our results remain the same using these alternative maturity cutoffs.

Third, we consider alternative filters on sample periods and bond funds. In [Appendix Table A7](#), we exclude the first two quarters of 2020; in [Appendix Table A8](#), we exclude non-U.S. funds; and in [Appendix Table A9](#), we exclude index funds. Our results remain robust using these alternative filters.

Fourth, we consider one alternative measure of bond fund ownership—the residual fund ownership from regressing on lagged Treasury return volatility. This is to tease out the component of lagged Treasury return volatility in determining fund ownership. Specifically, in each quarter, we conduct a cross-sectional regression of *Ownership* on the past four-quarter average Treasury

excess return volatility and obtain the residuals as an alternative measure of fund ownership, denoted *Ownership Residual*. *Ownership (High LMI) Residual* and *Ownership (Low LMI) Residual* are computed in the same way. Then, we reconduct our analyses in Tables 3 and 5. As shown in Appendix Table A10, we find that our main results remain unchanged using residual fund ownership.

Fifth, we also consider an alternative way to decompose bond fund ownership to provide additional evidence of the liquidity management mechanism. Specifically, we construct a fund-level measure of liquidity mismatch, which equals the difference between the total corporate bond portfolio weight minus the sum of portfolio weights of cash and government bond holdings.²⁴ This measure considers both the illiquid and cash-like assets in a fund's holdings.²⁵ Intuitively, we expect funds with severe liquidity mismatch to have a strong motive to conduct liquidity management, and thus, they exert a stronger effect on Treasury return volatility. To test this conjecture, we divide all bond funds at the end of each quarter into two groups: those with liquidity mismatch in the top tercile are categorized in the high-liquidity-mismatch group, while the rest are included in the low-liquidity-mismatch group. Afterward, for each U.S. Treasury at each quarter, we sum its bond fund ownership separately for the two groups of bond funds and denote them as *Ownership (High Liquidity Mismatch)* and *Ownership (Low Liquidity Mismatch)*. Finally, we conduct the same Fama-MacBeth regressions as those in Table 3 but replace *Ownership* with *Ownership (High Liquidity Mismatch)* and *Ownership (Low Liquidity Mismatch)*. Results in Appendix Table A11 confirm our conjecture: the positive association between bond fund ownership and Treasury return volatility is indeed stronger for Treasuries heavily held by bond funds with severe liquidity mismatch.

Finally, one might be concerned if the differential effect on volatility between low- and high-LMI fund ownership were driven by the low and high Treasury bond holdings by bond funds. That is, because low-LMI funds may not hold many Treasury bonds, they do not trade actively in the Treasury bond market. We rule out this possibility by decomposing bond fund ownership into two components: *Ownership (High Treasury Holding)* is the proportion of total market value of a Treasury that is held by funds with high Treasury holdings; and *Ownership (Low Treasury Holding)* is the proportion of total market value of a Treasury that is held by the rest of the funds. Treasury holding is defined as the total U.S. Treasury asset weight for each fund in each quarter. We rank all bond funds in each quarter, and the top tercile funds in each quarter are identified as funds with high Treasury holdings. Panel A of Appendix Table A12 shows that these two components of fund ownership have impacts on

²⁴ We identify asset types using detailed holding type ID provided by Morningstar. See Sections 1.2 and 1.4 for details.

²⁵ We thank the two anonymous reviewers for their great suggestions on this test.

Treasury volatility that are similar in magnitude, different from what we have documented in Table 5 for fund ownership split by *LMI*. Panel B of Appendix Table A12 shows that controlling for *Ownership (High Treasury Holding)* or *Ownership (Low Treasury Holding)* does not alter our main results.

2.4 A quasi-natural experiment: The Liquidity Risk Management Rule in 2017

While the results described in Sections 2.1, 2.2, and 2.3 are consistent with our argument that liquidity management induces excess volatility in Treasuries, we are aware of potential endogeneity issues. For example, it could be possible that bond funds prefer to hold Treasuries with certain characteristics that could lead to higher excess return volatility in the future. Thus, providing evidence for causal interpretation is critical to establishing our proposed liquidity management mechanism. The ideal method of identification is to find direct and exogenous shocks to the liquidity management of bond funds. Therefore, in this subsection, we exploit one recent policy—the introduction of the Liquidity Risk Management Rule (Rule 22e-4) in 2017 by the SEC—as a quasi-natural experiment that has had a direct effect on the liquidity management of bond funds.

The Liquidity Risk Management Rule was introduced on October 13, 2016 to promote effective liquidity risk management for U.S. open-end mutual funds. Under this regulation, U.S. funds are required to set up a minimum percentage of net assets that must be invested in highly liquid investments, such as U.S. Treasuries.²⁶ The rule went into effect on January 17, 2017; we expect U.S. mutual funds to have adjusted their liquidity management from the effective date.

We first empirically examine whether the introduction of the Liquidity Risk Management Rule affected the liquidity management of some U.S. bond funds. Intuitively, the Liquidity Risk Management Rule requires all U.S. bond funds to implement liquidity management using liquid assets (e.g., Treasuries). Thus, U.S. bond funds that conducted little liquidity management prior to this rule are strongly affected by it. In this sense, we expect Treasuries' trade-to-flow sensitivity for low-LMI U.S. funds to increase significantly after the implementation of the Rule. On the contrary, the trade-to-flow sensitivity of Treasuries for other funds should not be significantly affected.

To test our argument, we categorize all bond funds in our sample based on LMI as follows. First, we take the average of LMI for each fund for the years 2012–2016 (i.e., the preevent period) and rank all funds based on average LMI. Similarly, we categorize funds within the bottom two-thirds as low-LMI funds and those in the top tercile as high-LMI funds. We expect low-LMI funds to increase their trade-to-flow sensitivity for Treasuries after the introduction of

²⁶ For more details, see “Investment Company Liquidity Risk Management Program Rules.” <https://www.sec.gov/divisions/investment/guidance/secg-liquidity>.

the Liquidity Risk Management Rule. Other funds in our sample (including high-LMI U.S. funds and non-U.S. funds, which are not regulated by the Liquidity Risk Management Rule) should not exhibit a significant change in their trade-to-flow sensitivity.

Thus, we run the following regression for the years 2012–2021, that is, 5 years before and after the introduction of the Liquidity Risk Management Rule for low-LMI U.S. funds and other funds separately:

$$\begin{aligned}
 & \text{Net Buy (US Treasuries)}_{f,q} \\
 &= \alpha + \beta_1 \cdot \text{Fund Flow}_{f,q} \cdot \text{After}_q + \beta_2 \cdot \text{Fund Flow}_{f,q} \\
 &+ \gamma_1 \cdot \text{Fund Return}_{f,q} \cdot \text{After}_q \\
 &+ \gamma_2 \cdot \text{Fund Return}_{f,q} + \gamma_3 \cdot X_{f,q-1} \cdot \text{After}_q + \gamma_4 \cdot X_{f,q-1} + \delta_q + \varepsilon_{f,q},
 \end{aligned} \tag{11}$$

where $\text{Net Buy (US Treasuries)}_{f,q}$ is fund f 's trading on U.S. Treasuries in quarter q ; $\text{Fund Flow}_{f,q}$ is fund f 's net flows in quarter q ; $\text{Fund Return}_{f,q}$ is the gross return of fund f in quarter q ; and After_q is a dummy variable that equals one for quarters in or after 2017, and zero otherwise. $X_{f,q-1}$ represents $\text{Fund Flow}_{f,q-1}$ and $\text{Fund Return}_{f,q-1}$. We include year-quarter fixed effects and cluster standard errors by funds. The coefficient on $\text{Fund Flow}_{f,q} \cdot \text{After}_q$ captures the increase in trade-to-flow sensitivity after the introduction of the Liquidity Risk Management Rule.

Table 6 confirms the effect of the Liquidity Risk Management Rule. Columns (1)–(2) show that the trade-to-flow sensitivity significantly increased for low-LMI U.S. funds after 2017. For other funds, the effect is insignificant, as shown in columns (3)–(4). This finding shows that this rule forced more funds to conduct liquidity management, which could have intensified the selling of Treasuries during the COVID-19 crisis. In Appendix Table A13, we take advantage of the global coverage of our bond-fund holding data and separate funds by U.S. versus non-U.S. funds. We confirm that the Liquidity Risk Management Rule in 2017 affected only the liquidity management of U.S. funds.

Next, we use this policy shock as a quasi-natural experiment to pinpoint liquidity management's causal impact on Treasuries' excess volatility. Specifically, we conduct a difference-in-differences analysis. In the DiD analysis, we first separate U.S. Treasuries into treatment and control groups. The treatment group consists of Treasuries held heavily by the low-LMI U.S. funds. Specifically, we calculate each Treasury's average ownership from low-LMI U.S. funds in the pre-event window (2012–2016). Treasuries within the top tercile of this average ownership are considered to be held heavily by the low-LMI U.S. funds, that is, the treatment group, whereas the control group consists of all other Treasuries. This is motivated by the finding in Table 6, that is, low-LMI U.S. funds tend to intensify their flow-induced trading in Treasuries after the introduction of the Liquidity Risk Management Rule, leading to

Table 6
Fund trading on U.S. Treasuries and fund flows: Liquidity Risk Management Rule

DepVar:	<i>NetBuy (US Treasuries)_{f,q}</i>			
	U.S. low-LMI funds		Other funds	
	(1)	(2)	(3)	(4)
<i>Fund Flow_{f,q} × After</i>	0.594*** (4.8)	0.613*** (4.6)	−0.053 (−0.8)	−0.053 (−0.7)
<i>Fund Flow_{f,q}</i>	0.679*** (8.6)	0.656*** (8.0)	1.635*** (30.7)	1.715*** (29.0)
<i>Fund Flow_{f,q−1} × After</i>		−0.027 (−0.2)		0.004 (0.1)
<i>Fund Flow_{f,q−1}</i>		0.020 (0.3)		−0.270*** (−5.8)
<i>Fund Return_{f,q} × After</i>	−0.916 (−1.0)	−0.526 (−0.6)	1.070*** (2.7)	1.359*** (3.4)
<i>Fund Return_{f,q}</i>	0.592 (0.8)	0.304 (0.4)	−0.675** (−2.1)	−0.740** (−2.3)
<i>Fund Return_{f,q−1} × After</i>		−0.909 (−0.8)		0.619 (1.5)
<i>Fund Return_{f,q−1}</i>		1.415* (1.7)		0.335 (1.1)
Year-quarter fixed effects	Yes	Yes	Yes	Yes
# of Obs	12,731	12,731	39,760	39,760
Adj <i>R</i> ²	0.046	0.046	0.091	0.093

This table reports the regression results of U.S. Treasury trading on fund flows for U.S. low-LMI funds and other funds around the introduction of the Liquidity Risk Management Rule. Bond funds are divided into U.S. and non-U.S. funds based on their domicile countries. We define funds with low liquidity management intensity as follows. First, for each fund, we compute its liquidity management intensity as the coefficient estimate from a univariate regression of *NetBuy (US Treasuries)* on *Fund Flow* using data from the past 12 quarters. We require at least five quarterly observations to estimate this coefficient. Second, we take the average of this intensity for each fund between 2012 to 2016 and rank all funds in the sample based on this average intensity. The bottom two-thirds of funds are identified as funds with low liquidity management intensity, while the other funds are identified as funds with high liquidity management intensity. *NetBuy (US Treasuries)* measures the percentage share change of a fund's total holdings in U.S. Treasuries relative to its beginning of the quarter holdings. *After* is a dummy variable that equals one from 2017 onwards, and zero otherwise. *Fund Flow* is the quarterly fund flows. *Fund Return* is the quarterly fund return. All variables are winsorized by quarter at the 5th and 95th percentiles. We control for year-quarter fixed effects and standard errors are clustered by funds. *t*-statistics are reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01. The sample period is from 2012 through 2021, that is, 5 years before and after the introduction of the Liquidity Risk Management Rule.

higher excess volatility in the Treasuries they hold. Thus, we run the following regression over the sample period, 2012 through 2021:

$$\begin{aligned} Volatility_{i,q} = & \alpha + \beta \cdot Treat_i \cdot After_q + \gamma \cdot Treat_i \\ & + \theta \cdot X_{i,q-1} + \phi \cdot X_{i,q-1} \cdot After_q + \varepsilon_{i,q}, \end{aligned} \tag{12}$$

where *Volatility_{i,q}* refers to the excess return volatility for U.S. Treasury *i* in quarter *q*; *Treat* is a dummy variable that equals one for the treated Treasuries, and zero otherwise; *After* is a dummy variable that equals one for quarters in or after 2017, and zero otherwise; and *X_{i,q−1}* represents the same set of control variables as in Table 3. We also include the interaction terms between each Treasury characteristic and the *After* dummy to control for potential time-varying effects on volatility driven by these characteristics after the introduction of the Liquidity Risk Management Rule. Based on the effect reported in Table 6, we conjecture that Treasuries held heavily by low-LMI U.S. funds should exhibit stronger excess return volatility after 2017.

Table 7 reports the results and confirms our conjecture. For example, column (6) shows that the coefficient on $Treat_i \cdot After_q$ is significantly positive (t -statistic = 2.7). In other words, Treasuries heavily held by low-LMI U.S. funds experienced an increase in excess volatility after the introduction of the Liquidity Risk Management Rule relative to the control group. Also, the estimation in column (6) suggests that the treatment variable after 2017 has a coefficient of 0.002 ($=0.003 - 0.001$) and is significant at 1% (F -test = 7.96). The coefficient before $Treat$ appears to be negative (t -stats. range from -0.7 to -2.7); this is because, by definition, the Treasuries in the treatment group were heavily held by bond funds with low liquidity management intensity before the event. Thus, their excess return volatility could be lower than that of the control group before 2017.

We have also examined parallel pre-trends and confirmed that the aforementioned results are not due to pre-trends before 2017 (see [Appendix Table A14](#)). In addition, according to [Park \(2021\)](#), though introduced in January 2017, the compliance date for this Liquidity Risk Management Rule was December 1, 2018, for large funds and June 1, 2019, for small funds. Thus, we should observe a gradual increase of the treatment effect. Considering this, in [Appendix Table A15](#), we conduct a quarter-by-quarter analysis of the treatment effect around the implementation window between 2017 to 2019. Indeed, the treatment effect only became statistically significant and economically meaningful since 2018Q2.

To corroborate the above-mentioned finding, we conduct a placebo test. Instead of focusing on ownership from low-LMI U.S. funds (which are heavily affected by the Liquidity Risk Management Rule), we examine whether Treasuries heavily held by non-U.S. funds, which are not affected by the Liquidity Risk Management Rule, experienced a similar effect after the introduction of the Rule. We expect to find non-results from this analysis. For this placebo test, the treatment group is the Treasuries, of which the average non-U.S. bond fund ownership from 2012 through 2016 is within the top tercile of our sample. The rest of the Treasuries are included in the control group. Then, we rerun the regression of Equation (12), and the results are reported in [Appendix Table A16](#). Indeed, Treasuries heavily held by non-U.S. bond funds were not affected by the introduction of the Liquidity Risk Management Rule, highlighting the effect of the Rule on low-LMI U.S. funds.

The results in this subsection not only pin down the causal impact of liquidity management on Treasury excess return volatility, but also have important policy implications. Our study suggests that the Liquidity Risk Management Rule has had unintended consequences and increases the Treasury market's fragility.²⁷

²⁷ A contemporaneous related work on the 2017 Liquidity Risk Management Rule is [Park \(2021\)](#). But [Park \(2021\)](#) focuses on the implications for the corporate bond market. He shows that this rule forces bond funds to hold more liquid corporate bonds and leads to more return comovement of corporate bonds, suggesting that the Liquidity Risk Management Rule has unintended consequences on the corporate bond market.

Table 7
Bond fund ownership and U.S. Treasury return volatility: Liquidity Risk Management Rule

DepVar:	Volatility (in %)					
	Market-adjusted returns			Three-factor-adjusted returns		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treat</i> × <i>After</i>	0.003** (2.3)	0.003*** (2.6)	0.002** (2.2)	0.003*** (2.7)	0.003*** (3.0)	0.003*** (2.7)
<i>Treat</i>	−0.002** (−2.3)	−0.002** (−2.5)	−0.000 (−0.7)	−0.002** (−2.5)	−0.002*** (−2.7)	−0.001 (−0.9)
<i>Time-to-maturity</i> × <i>After</i>	−0.002*** (−13.9)	−0.002*** (−13.7)	−0.002*** (−9.9)	−0.001*** (−13.2)	−0.001*** (−12.9)	−0.001*** (−9.2)
<i>Time-to-maturity</i>	0.008*** (106.3)	0.008*** (102.6)	0.008*** (84.0)	0.007*** (102.7)	0.007*** (98.8)	0.007*** (81.6)
<i>Coupon Rate</i> × <i>After</i>		0.000 (0.7)	−0.001*** (−2.6)		0.000 (0.6)	−0.001*** (−2.9)
<i>Coupon Rate</i>		−0.001*** (−7.4)	−0.000 (−1.5)		−0.001*** (−6.3)	−0.000 (−0.3)
<i>On-the-run</i> × <i>After</i>		0.004** (2.1)	0.001 (0.7)		0.004** (2.0)	0.001 (0.7)
<i>On-the-run</i>		−0.003* (−1.9)	−0.001 (−0.7)		−0.002* (−1.7)	−0.001 (−0.4)
<i>Log(Size)</i> × <i>After</i>			−0.002 (−1.5)			−0.002* (−1.8)
<i>Log(Size)</i>			0.004*** (3.7)			0.004*** (3.9)
<i>Bid-ask Spread</i> × <i>After</i>			−0.121** (−2.3)			−0.135*** (−2.9)
<i>Bid-ask Spread</i>			−0.304*** (−10.4)			−0.264*** (−9.8)
Year-quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs	11,067	11,067	11,067	11,067	11,067	11,067
Adj <i>R</i> ²	0.845	0.846	0.851	0.843	0.844	0.849

This table reports regression results of U.S. Treasury excess return volatility on bond fund ownership around the introduction of the Liquidity Risk Management Rule. We classify Treasuries into treatment group and control group as follows. First, for each fund, we compute its LMI as the coefficient estimate from a univariate regression of *NetBuy (US Treasuries)* on *Fund Flow* using data from the past 12 quarters, where *NetBuy (US Treasuries)* is the percentage change of a fund's total holdings in U.S. Treasuries relative to its beginning of the quarter holdings, and *Fund Flow* is the quarterly fund flows. We require at least five quarterly observations to estimate this coefficient. Second, we take the average of this intensity for each fund between 2012 to 2016 and rank all funds in the sample based on this average intensity. The bottom two-thirds of funds are identified as low-LMI funds, while the rest of the funds are identified as high-LMI funds. Third, we aggregate U.S. bond fund ownership from funds with low-LMI for each Treasury in each quarter and compute the average of this ownership between 2012 to 2016. Treasuries are ranked based on this average low-LMI U.S. fund ownership. *Treat* is a dummy variable that equals one for Treasuries within the top tercile, and zero otherwise. *Volatility (in %)* is the standard deviation (in percentage) of the daily risk-adjusted returns in a quarter. We consider two ways to compute risk-adjusted returns: market-adjusted and three-factor-adjusted. Market-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market and its two lags; three-factor-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market, investment-grade bonds, junk bonds, and their two lags. *After* is a dummy variable that equals one from 2017 onwards, and zero otherwise. *Time-to-maturity* is the years between the quarter-end and maturity date. *Coupon Rate* is the coupon rate expressed as a percentage. *On-the-run* is a dummy variable that equals one if a Treasury is the most recently issued Treasury of a particular maturity, and zero otherwise. *Log(Size)* is the logarithm of the total amount outstanding (face value, in millions of USD). *Bid-ask Spread* is the difference between ask price and bid price. All independent variables (except for the *On-the-run* dummy) are winsorized at the 1st and 99th percentiles. We control for year-quarter fixed effects, and *t*-statistics based on robust standard errors are reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01. The sample period is from 2012 through 2021, that is, 5 years before and after the introduction of the Liquidity Risk Management Rule.

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2.5 The COVID-19 Treasury market turmoil

To provide additional evidence for the causal interpretation, we exploit the COVID-19 Treasury market turmoil. In this subsection, we extend our study and examine whether the economic mechanism of liquidity management contributed to the Treasury market turmoil during the COVID-19 crisis. The outbreak of COVID-19 in the United States induced significant outflows from bond mutual funds. Based on our proposed mechanism, Treasuries heavily held by bond funds with high liquidity management intensity should experience dramatic price declines during the crisis. Furthermore, since the price declines were driven by collective fire sales, the prices should subsequently reverse after the intervention of the Federal Reserve.

On March 11, 2020, the World Health Organization announced that COVID-19 had become a global pandemic.²⁸ As the outbreaks in the United States and other countries brought unprecedented uncertainty to the global economy, bond funds started to experience a large number of outflows from the second week of March 2020. On March 23, the Federal Reserve announced a series of extensive, unprecedented measures to stabilize markets and support the economy; this effectively calmed panicked investors and reversed the trend of redemptions within a few weeks. As shown in Figure 1, panel A, the average daily fund flow decreased sharply following the pandemic announcement and slowly reverted back after the Fed's intervention on March 23. Figure 1, panel B shows that the aggregate capital outflow from the bond funds in our sample between March 11 and March 23 was about 5.07% of their total net assets at the beginning of the year. This pattern is also documented in detail by [Falato, Goldstein, and Hortaçsu \(2021\)](#).

To formally examine the association between fund ownership and the fragility of U.S. Treasury prices during the COVID-19 crisis, we conduct difference-in-differences analyses in two event windows. The first event window consists of the 2 weeks before and after the pandemic announcement on March 11. Immediately following the pandemic announcement, the market witnessed full-blown liquidity turmoil, as bond mutual funds faced soaring redemptions and were forced to sell their holdings. Under the liquidity management mechanism, we expect strong price declines for U.S. Treasuries heavily held by bond funds with high LMI. Given the short span of the crisis, we focus on a panel of daily observations from February 24, 2020, to March 20, 2020. Specifically, we follow [Ma, Xiao, and Zeng \(2022\)](#) and run the following panel regression:

$$\begin{aligned} \text{Daily Return}_{i,t} = & \alpha + \beta_1 \cdot \text{High Ownership (High LMI)}_i \cdot \text{After}_t \\ & + \beta_2 \cdot \text{High Ownership (Low LMI)}_i \cdot \text{After}_t \quad (13) \\ & + \delta_i \cdot \phi_t + \psi_i + \varepsilon_{i,t}. \end{aligned}$$

²⁸ See WHO Director-General's opening remarks at the media briefing on COVID-19 on March 11, 2020. <https://www.who.int/director-general/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19-11-march-2020>.

Table 8
Ownership and U.S. Treasury returns around the COVID-19 crisis and the Fed intervention

DepVar:	Daily Return (in basis points)			
	Around the COVID-19 crisis		Around the Fed's intervention	
	(1)	(2)	(3)	(4)
<i>High Ownership (High LMI) × After</i>	−20.347*** (−3.2)	−4.654*** (−2.7)	12.574*** (3.3)	4.039*** (2.9)
<i>High Ownership (Low LMI) × After</i>	−1.362 (−0.3)	1.574 (0.9)	−0.916 (−0.3)	−1.292 (−0.9)
<i>After</i>	−32.000*** (−10.0)		9.346*** (5.3)	
Maturity-day fixed effects	No	Yes	No	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
# of Obs	5,960	5,960	5,960	5,960
Adj R^2	0.030	0.945	0.004	0.940

This table reports regression results of U.S. Treasury daily returns on bond fund ownership around the COVID-19 crisis and the Fed announcement. The dependent variable is a Treasury's daily return, expressed in basis points. In the first two columns, the sample period spans from February 24th to March 20th in 2020. In the last two columns, the sample period spans from March 9th to April 3rd in 2020. We decompose bond fund ownership into two components: *Ownership (High LMI)* is the proportion of total market value of a Treasury that is held by high-LMI funds; *Ownership (Low LMI)* is the proportion of total market value of a Treasury that is held by the rest of the funds. Liquidity management intensity (LMI) is defined as the coefficient estimate from a univariate regression of *NetBuy (US Treasuries)* on *Fund Flow* using data from the past 12 quarters, where *NetBuy (US Treasuries)* is the percentage change of a fund's total holdings in U.S. Treasuries relative to its beginning of the quarter holdings, and *Fund Flow* is the quarterly fund flows. We require at least five quarterly observations to estimate this coefficient. The top one-third of funds in each quarter are identified as funds with high-LMI based on this estimate. *High Ownership (High LMI)* is a dummy variable that equals one for Treasuries with *Ownership (High LMI)* in the top tercile. *High Ownership (Low LMI)* is a dummy variable that equals one for Treasuries with *Ownership (Low LMI)* in the top tercile. *After* is a dummy variable that equals one for the 2 weeks of crisis period in the first two columns, and the 2 weeks following the Fed intervention in the last two columns. We divide all Treasuries into six maturity buckets (less than 2 years, 2–4 years, 4–6 years, 6–8 years, 8–10 years, 10+ years) and control for maturity-day fixed effects, as well as bond fixed effects. Standard errors are clustered by Treasuries. t -statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

The dependent variable is daily U.S. Treasury returns (in basis points). The key independent variable is *High Ownership (High LMI)_{*i*}*, a dummy variable that equals one for U.S. Treasuries with ownership from high-LMI funds in the top tercile. We also include another dummy variable *High Ownership (Low LMI)_{*i*}*, that equals one for U.S. Treasuries with ownership from low-LMI funds in the top tercile. All ownership variables are constructed using bond fund holdings at the end of 2019Q4. *After_{*t*}* is a dummy variable that equals one for days on or after March 11, 2020, and zero otherwise. We control for maturity-day fixed effects ($\delta_i \cdot \phi_t$) and bond fixed effects (ψ_i), and cluster standard errors by Treasuries.²⁹

Table 8, columns (1)–(2) report results consistent with our conjecture. As shown in column (2), where we control both maturity-day fixed effects and bond fixed effects, we find that U.S. Treasuries heavily held by high-LMI funds experienced a 4.654 bps (t -statistic = −2.7) decrease per day in price during the 2 weeks following the pandemic announcement. In contrast, U.S. Treasuries

²⁹ For the maturity-day fixed effects, we divide all Treasuries into six maturity buckets: less than 2 years, 2–4 years, 4–6 years, 6–8 years, 8–10 years, and more than 10 years.

heavily held by low-LMI bond funds did not experience significant price drops after the pandemic announcement.

To illustrate that the price impacts shown in columns (1)–(2) of Table 8 were nonfundamental, we conduct the second event study. The event windows consist of 2 weeks before and after the start of the Fed's intervention on March 23. This episode provides an ideal setting to study whether the price declines on Treasuries heavily held by high-LMI funds rose temporarily from nonfundamental sales.

We follow the same empirical specification as in Equation (13) but use the sample period from March 9 to April 3 in 2020, that is, 2 weeks before and after the Fed intervened. All variables are the same, except that $After_t$ now equals one for days on or after March 23, 2020, and zero otherwise. The results reported in Table 8, columns (3)–(4), confirm our conjecture. Treasuries heavily held by high-LMI funds experienced 4.039 bps price increases per day during the 2 weeks following the Fed's intervention (t -statistic = 2.9). In other words, the price decline during the COVID-19 period was fully reversed, consistent with our price pressure argument due to liquidity management.

Overall, the patterns documented above are consistent with our argument that the liquidity management with Treasuries contributed to the U.S. Treasury market turmoil during the COVID-19 pandemic in March 2020. The results in this subsection, together with those in Section 2.4, can help explain why the price reaction of Treasuries to the COVID-19 crisis did not follow the established crisis playbook. As we know, during periods of financial market turmoil, the prices of U.S. Treasuries (the safe assets in the conventional view) usually increase due to flight-to-safety and flight-to-liquidity. The COVID-19 crisis, however, differed from previous crises, and Treasuries experienced sharp price declines. This phenomenon is puzzling to academics and practitioners and raises concerns about the safe haven status of U.S. Treasuries. Our results indicate that the COVID-19 Treasury market turmoil was (at least partially) an unintended consequence of the 2017 Liquidity Risk Management Rule. As we show, the Rule forced many bond funds to conduct liquidity management with liquid assets, which in turn intensified the selling of Treasuries in the presence of unprecedented outflows at the beginning of the crisis. In this sense, we provide novel insights into the puzzling phenomenon during the COVID-19 period and contribute to a growing body of literature on this topic (see, e.g., Duffie 2020; He, Nagel, and Song 2022).³⁰

Meanwhile, we do not intend to claim that this is the only mechanism that drove the COVID-19 episode; several concurrent studies propose other

³⁰ The study by Ma, Xiao, and Zeng (2022) does not examine the role of the Liquidity Risk Management Rule on the COVID-19 Treasury turmoil. Also, the sample period of our study is different from that of Ma, Xiao, and Zeng (2022); we focus on a much longer sample period that covers both normal and crisis times. In contrast, Ma, Xiao, and Zeng (2022) focus only on the COVID-19 turmoil (February to March 2020). Our longer sample period allows for generalizing findings beyond extreme market conditions and, as a result, helps avoid confounding factors during the COVID-19 period.

mechanisms that could have caused this turmoil in the Treasury market (see, e.g., [Duffie 2020](#); [Fleming and Ruela 2020](#); [Schrimpf, Shin, and Sushko 2020](#); [Kruttli et al. 2021](#); [He, Nagel, and Song 2022](#)). These channels, including ours, are not exclusive, and more granular data is needed to quantify each channel's contribution to the turmoil in the Treasury market.

3. Additional Tests and Further Discussion

3.1 Common ownership and return comovement

To shed further light on the price fragility in the Treasury market, we examine the return comovement of U.S. Treasuries. Return correlations among individual assets within a particular asset class largely determine the total return variance of the asset class, a conventional measure of fragility. With nonfundamental flow-induced trading, the prices of Treasuries commonly held by bond funds tend to comove due to their systemic exposure to fund flow shocks.

Our argument is not new in the literature, but there are some novel empirical predictions in the context of liquidity management of bond mutual funds. [Greenwood and Thesmar \(2011\)](#) and [Anton and Polk \(2014\)](#) study the association between the ownership of equity mutual funds and stock return comovement. They find that stocks commonly held by mutual funds tend to comove in prices due to correlated fund trading. While the underlying mechanism of our paper is similar to these existing works, our focus on bond funds and their liquidity management leads to several unique predictions for Treasuries. First, as discussed earlier, since bond funds trade Treasuries aggressively to accommodate fund flows, Treasuries with high common ownership should exhibit stronger excess return comovements. Second, this effect should be stronger during market downturns since liquidity management becomes more urgent when funds face outflows. Such excessive downside market comovement can be considered an indicator of the increased systematic risk in Treasury prices during market downturns.

To test our argument, we follow [Anton and Polk \(2014\)](#) and run [Fama and MacBeth \(1973\)](#) regressions to examine the effect of common ownership on the Treasury return comovement.³¹ The regression specification is as follows,

$$Corr_{i,j,q} = \alpha + \beta \cdot \text{Common Ownership}_{i,j,q-1} + \theta \cdot X_{i,j,q-1} + \varepsilon_{i,j,q}, \quad (14)$$

where $Corr_{i,j,q}$ is the excess return comovement between bonds i and j in quarter q , and the key independent variable is $\text{Common Ownership}_{i,j,q-1}$,

³¹ Alternatively, we could follow [Greenwood and Thesmar \(2011\)](#) to estimate how the flow-induced trading of mutual funds affects asset return comovement. The approach from [Greenwood and Thesmar \(2011\)](#) depends on the structure of fund flows (e.g., the variance-covariance matrix of fund flows among different funds). Our analysis, which follows [Anton and Polk \(2014\)](#), does not involve stylized assumptions on parameter values.

which measures the extent to which bonds i and j are held by the same bond funds. Following Anton and Polk (2014), we control for the Treasury pair's similarities in bond characteristics ($X_{i,j,q-1}$): *Time-to-maturity Difference* is the absolute difference between the years to maturity for two Treasuries; *Coupon Rate Difference* is the absolute difference between the coupon rates of two Treasuries; *On-the-run Difference* is the absolute difference in the on-the-run status, which describes whether a Treasury is the most recently issued of a particular maturity; *Log(Size) Difference* is the absolute difference between two Treasuries' logarithm of total amount outstanding; *Bid-ask Spread Difference* is the absolute difference between two Treasuries' bid-ask spread. We compute Newey and West (1987) standard errors corrected by serial dependence of three lags.

Table 9 reports the results. As shown in columns (1) and (5), common ownership positively forecasts comovement among Treasuries. For example, as shown in column (1), the coefficient on *Common Ownership* is 3.252 (t -statistic = 10.7) after including all control variables. This suggests that a 1% increase in *Common Ownership* is associated with a 3.252% increase in the average market-adjusted pairwise correlation between two Treasuries. The effect we document here is also economically meaningful, considering that the average of the market-adjusted excess return comovement among Treasuries is 15.3%.

Next, we turn to test the asymmetric pattern, that is, the effect of common ownership on the Treasury return comovement should be stronger during market downturns. The prediction is motivated by the fact that when the Treasury market declines, bond funds experience fund outflows (see, Brooks, Katz, and Lustig 2020), and liquidity management is more urgent, leading to a stronger association between common ownership and return comovement in Treasuries. This test is also related to common observations that return covariance plays a more important role in the total return variance of the asset class during periods of market turmoil, compared to normal periods.

We measure this asymmetry in return comovement with the following steps. Within each quarter, we first sort all trading days into two equal groups (downside markets and upside markets) based on the daily aggregate Treasury market returns. Then, we calculate the return comovement for each group, denoted *Corr_down* and *Corr_up*, respectively. Finally, we define *Down-minus-up* as the difference between *Corr_down* and *Corr_up*. Note that, since *Down-minus-up* is based on the same pair of Treasuries, this asymmetry measure has a unique advantage in eliminating potential similarities in unobservable bond characteristics that may drive return comovement.

We then replace the dependent variable from Equation (14) and run Fama-MacBeth regressions of *Corr_down*, *Corr_up*, and *Down-minus-up* on common ownership, respectively, to examine the asymmetric effect of common ownership on Treasuries comovement between downside and upside markets. Table 9, column (4) confirms our conjecture. A 1% increase in *Common Ownership* is associated with a 0.188% (t -statistic = 2.4) increase in

Table 9
Common ownership and U.S. Treasury return comovement

DepVar:	Market-adjusted returns				Three-factor-adjusted returns			
	<i>Corr_all</i>	<i>Corr_down</i>	<i>Corr_up</i>	<i>Down-minus-up</i>	<i>Corr_all</i>	<i>Corr_down</i>	<i>Corr_up</i>	<i>Down-minus-up</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Common Ownership</i>	3.252*** (10.7)	3.338*** (11.3)	3.150*** (10.0)	0.188** (2.4)	3.224*** (10.4)	3.317*** (10.9)	3.128*** (9.8)	0.188** (2.3)
<i>Time-to-maturity Difference</i>	-0.052*** (-35.0)	-0.052*** (-35.0)	-0.052*** (-33.0)	-0.000 (-0.1)	-0.051*** (-31.7)	-0.051*** (-31.1)	-0.051*** (-29.9)	-0.000 (-0.5)
<i>Coupon Rate Difference</i>	-0.025*** (-5.5)	-0.024*** (-5.2)	-0.026*** (-5.7)	0.001 (1.0)	-0.025*** (-5.5)	-0.024*** (-5.1)	-0.026*** (-5.8)	0.002 (1.6)
<i>On-the-run Difference</i>	0.031*** (5.0)	0.029*** (4.6)	0.031*** (5.3)	-0.002 (-1.2)	0.031*** (5.0)	0.030*** (4.8)	0.031*** (5.1)	-0.001 (-0.9)
<i>Log(Size) Difference</i>	-0.006 (-0.4)	-0.004 (-0.3)	-0.007 (-0.5)	0.003 (0.9)	-0.005 (-0.3)	-0.003 (-0.2)	-0.006 (-0.4)	0.003 (0.8)
<i>Bid-ask Spread Difference</i>	-4.836*** (-4.6)	-4.909*** (-4.7)	-4.734*** (-4.6)	-0.175 (-0.6)	-4.735*** (-4.6)	-4.842*** (-4.6)	-4.585*** (-4.5)	-0.257 (-1.1)
# of Obs	2,040,889	2,040,889	2,040,889	2,040,889	2,040,889	2,040,889	2,040,889	2,040,889

This table reports the results from Fama-MacBeth regressions of U.S. Treasury excess return comovement in quarter q on common ownership in quarter $q - 1$. *Corr_all* is the excess return correlation between two securities in a quarter. The excess return correlation is computed as the pairwise correlation of daily risk-adjusted returns for a pair of securities. We consider two ways to compute risk-adjusted returns: market-adjusted and three-factor-adjusted. Market-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market and its two lags; three-factor-adjusted daily returns are obtained as the residuals from a regression of daily bond excess return on returns from the aggregate Treasury market, investment-grade bonds, junk bonds, and their two lags. We further sort all trading days in a quarter into two equal groups (downside and upside market days) based on the aggregate Treasury market returns, and *Corr_down* and *Corr_up* are the excess return correlation between two securities during downside and upside market days, respectively. *Down-minus-up* is the difference between *Corr_down* and *Corr_up*. All these pairwise correlations are constructed using market-adjusted returns and three-factor-adjusted returns. *Common Ownership* is the proportion of total market value of a Treasury pair held by funds holding the pair of Treasuries. *Time-to-maturity Difference* is the absolute difference between two securities' years-to-maturity. *Coupon Rate Difference* is the absolute difference between two securities' coupon rates. *On-the-run Difference* is the absolute difference between two Treasuries' on-the-run status for a pair of Treasuries, where *On-the-run* is a dummy variable that equals one if a Treasury is the most recently issued Treasury of a particular maturity, and zero otherwise. *Log(Size) Difference* is the absolute difference between two Treasuries' logarithm of total amount outstanding. *Bid-ask Spread Difference* is the absolute difference between two Treasuries' bid-ask spread. All independent variables (except for the *On-the-Run Difference* dummy) are winsorized at the 1st and 99th percentiles. Heteroscedasticity and auto-correlation-consistent Newey-West (1987) t -statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. The sample period is from 2002Q4 through 2021Q4.

market-adjusted *Down-minus-up*. In other words, Treasury pairwise correlation becomes significantly higher during downside markets relative to upside markets.

For comparison, we repeat the same exercises on corporate bonds. We expect that the association between fund ownership and the return comovement should be weaker for corporate bonds, as funds tend to avoid selling corporate bonds immediately to accommodate redemptions. [Appendix Table A17](#) reports the results, and we obtain two findings. First, while *Common Ownership* can also significantly forecast the excess return comovement on corporate bonds, the economic magnitude is much smaller than that on Treasuries. More importantly, the asymmetric effect of *Common Ownership* on the return comovement between downside and upside markets for corporate bonds is insignificant. This pattern is consistent with the results, reported in [Table 2](#), that corporate bonds are less sensitive to flow shocks, as bond funds tend to avoid trading corporate bonds (e.g., due to high price impacts) to meet liquidity needs.

Overall, the analyses based on return comovement enrich our understanding of the relationship between bond fund ownership and U.S. Treasury price fragility, and highlight the unique implications of using U.S. Treasuries in the liquidity management practice.

3.2 The establishment of the Secondary Market Corporate Credit Facility

While the results described in [Section 2.4](#) uncover an unintended consequence of the 2017 Liquidity Risk Management Rule, it is important to understand which types of policies can mitigate fragility in the Treasury and bond markets. Among the various Fed interventions during the COVID-19 crisis, one was of particular interest to the liquidity management mechanism: the establishment of the Secondary Market Corporate Credit Facility. As the Fed's first ever intervention in corporate bond markets, the SMCCF was introduced to address the severe dislocations in the corporate bond market caused by the COVID-19 pandemic and associated economic shutdowns. The goal of the facility was to support market liquidity and the availability of credit for large corporations; it provided price support to U.S. investment-grade corporate debts with a time-to-maturity of less than 5 years.³²

The SMCCF was of particular importance to the liquidity management mechanism we analyze. Unlike the Liquidity Risk Management Rule, which requires liquidity management, the SMCCF adopted an alternative approach to manage liquidity risk of bond funds by providing direct liquidity to illiquid assets. Since the SMCCF provided market liquidity to corporate bonds (see evidence in [Li and Ringgenberg 2023](#)), it reduced the need for liquidity

³² For more detailed discussions on the SMCCF, see [Boyarchenko et al. \(2022\)](#) and [Clarida, Duygan-Bump, and Scotti \(2021\)](#).

management for bond funds that held significant amounts of SMCCF-eligible corporate bonds. In this sense, we argue that Treasuries heavily held by bond funds with high exposure to SMCCF-eligible bonds should experience less excess return volatility following the commencement of the SMCCF.

To examine this conjecture, we construct a measure of funds' exposure to the SMCCF-eligible bonds using the following steps. First, for each fund in 2019Q4, we compute the portfolio weight of U.S. investment-grade corporate bonds with dates of maturity before September 2025 (i.e., they mature within 5 years of the end of 2020Q2). We rank all funds based on this portfolio weight. Then, we classify funds within the top tercile as those with high exposure to the SMCCF and classify the rest as funds with low exposure. Finally, for each Treasury at 2019Q4, we calculate its fund ownership separately for bond funds with high and low SMCCF exposure; these are denoted as *Ownership (High SMCCF Exposure)* and *Ownership (Low SMCCF Exposure)*, respectively. Due to the nature of this event, we focus on a short sample period from 2020Q1 through 2020Q2 and run the following OLS regression:

$$\begin{aligned}
 Volatility_{i,t} = & \alpha + \beta_1 \cdot Ownership(High\ SMCCF\ Exposure)_i \times After_t \\
 & + \beta_2 \cdot Ownership(Low\ SMCCF\ Exposure)_i \times After_t \\
 & + \gamma_1 \cdot Ownership(High\ SMCCF\ Exposure)_i \\
 & + \gamma_2 \cdot Ownership(Low\ SMCCF\ Exposure)_i \\
 & + \gamma_3 \cdot After_t + \phi \cdot X_i + \varepsilon_{i,t}.
 \end{aligned} \tag{15}$$

The dependent variable is the excess return volatility for U.S. Treasuries, computed from the beginning of 2020 through March 20, 2020, and from March 23, 2020, through the end of the second quarter of 2020. *After_t* is a dummy variable that equals one for dates on or after March 23, 2020, and zero otherwise. We control for time-to-maturity, coupon rate, and on-the-run status. Empirical results are consistent with our conjecture: the estimation of β_1 appears significantly negative. Treasuries heavily held by bond funds with high exposure to the SMCCF indeed experienced lower excess return volatility after the SMCCF (see [Appendix Table A18](#)).

In sum, the results are consistent with the intuition that when corporate bonds in the portfolios of bond funds become more liquid, funds are less concerned about liquidity issues and thus conduct less liquidity management with Treasuries. Consequently, bond funds exert less fragility in Treasuries; this strengthens our argument as to the role of liquidity management in Treasury fragility.

The results here, together with the analysis in Section 2.4, have important policy implications. We show evidence that the Liquidity Risk Management Rule can have unintended consequences for the volatility of the Treasury market. Also, [Park \(2021\)](#) finds that this rule induces more return comovement

of corporate bonds. Our extension in this section suggests that enhancing corporate bond liquidity instead of the liquidity management policy can not only help the corporate bond market (see [Li and Ringgenberg 2023](#)) but can also potentially mitigate fragility in the Treasury market.

4. Conclusion

In recent years, the U.S. Treasury market—which used to be considered the world’s most liquid market—has become more fragile, as seen in the “flash rally” episode in 2014 and the turmoil during the outbreak of COVID-19. Given the importance of Treasuries in the global financial system, it is necessary to understand the underlying economic mechanism through which the fragility occurs.

We empirically test our argument that the liquidity management of open-end bond mutual funds can transmit non-fundamental demand shocks from fund flows into Treasuries and can lead to Treasury fragility. We have several empirical findings to support our argument. First, we document that bond funds aggressively trade Treasuries to manage their liquidity needs, as the trade-to-flow sensitivity is larger on Treasuries than on other government bonds and corporate bonds. Second, we conduct cross-sectional studies and find that Treasuries’ excess return volatility—our primary measure of fragility—is positively associated with bond fund ownership. We further construct a fund-level measure of liquidity management intensity and show that the positive association between Treasury excess return volatility and fund ownership is much stronger for funds with high liquidity management intensity. Third, we exploit one recent policy, the SEC’s Liquidity Risk Management Rule in 2017, as a shock to funds’ liquidity management; we do so to pin down the causal impact of the liquidity management of bond funds on Treasury excess return volatility. Finally, we find evidence that liquidity management with Treasuries contributes at least partially to the recent COVID-19 Treasury market turmoil. That is, Treasuries heavily held by funds with high liquidity management intensity experienced large price declines during the COVID crisis, followed by subsequent reversals.

In sum, our study provides direct and causal evidence on the effect of open-end bond funds’ liquidity management on Treasury fragility. We contribute to two strands of literature. First, our study contributes to the literature on liquidity management and its asset pricing implications. To the best of our knowledge, we are the first to use the 2017 Liquidity Risk Management Rule as an exogenous shock to the liquidity management of open-end bond funds with Treasuries and to provide causal evidence of the effect of liquidity management on Treasury fragility. Second, our paper contributes to the growing number of studies on the recent COVID-19 Treasury market turmoil. It suggests that the 2017 Liquidity Risk Management Rule and the increasing size of bond funds contributed to the turmoil during the COVID-19 period. Our study has

important policy implications and cautions against the possible unintended consequences of liquidity management policies.

Code Availability. The replication code is available in the Harvard Data-verse at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi%3A10.7910%2FDVN%2FH6E25Z&version=DRAFT>.

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